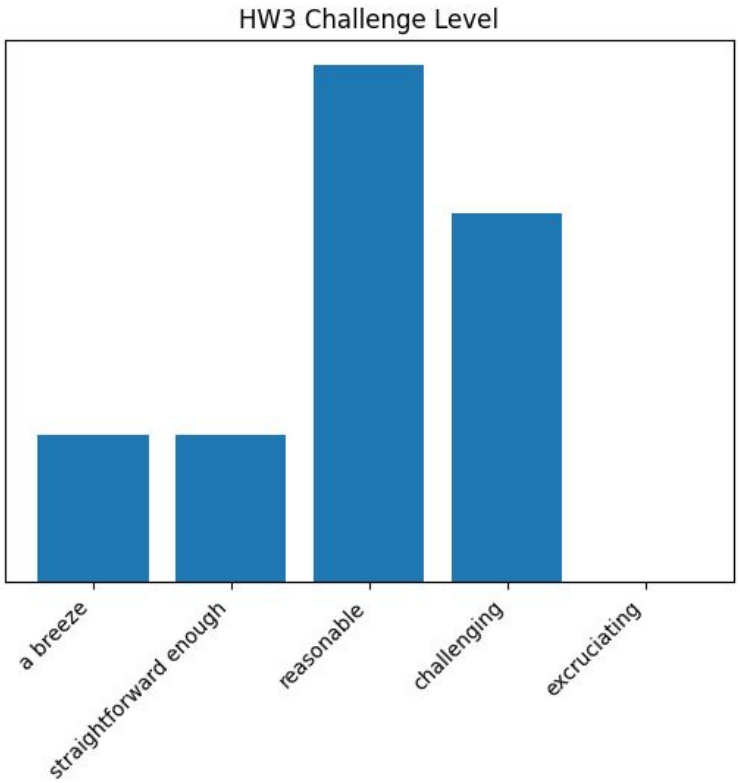
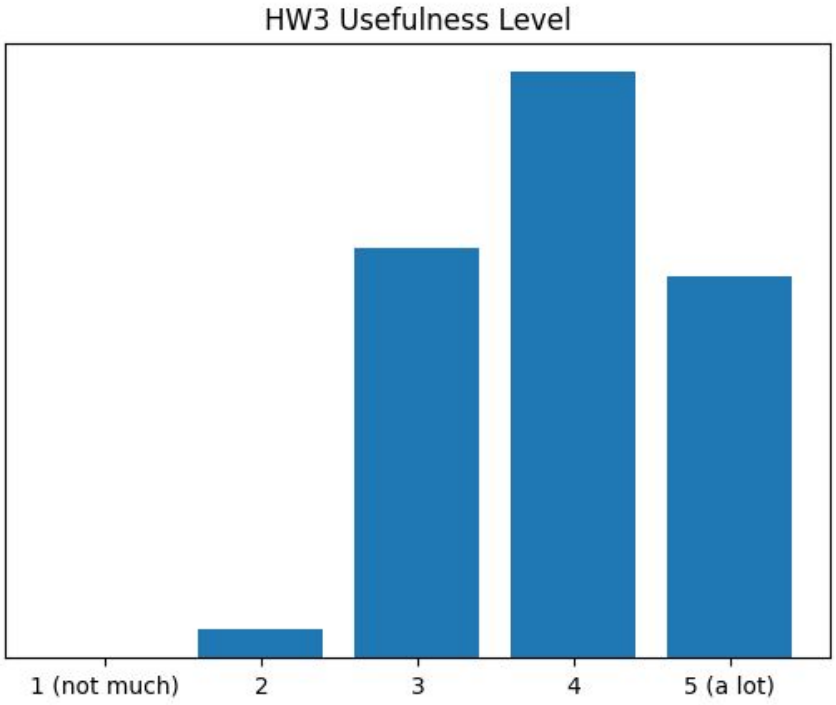
CSE 493L Section 5

Data Structures in Rust

Looking Back on Homework 3





Homework 5: Due Dates

- **Monday, February 9th:** test case - hard deadline
- **Wednesday, February 11th**: code, questions - 2 day grace period

Implementing single-threaded `insert` and `get` on a Red-Black Tree!

Check-in

- Before we get started: *any questions?*

Also make sure everyone has a handout!

Aside: Packed vs Linked Data Structures

Packed data structures *tend to be** more performant than linked ones.

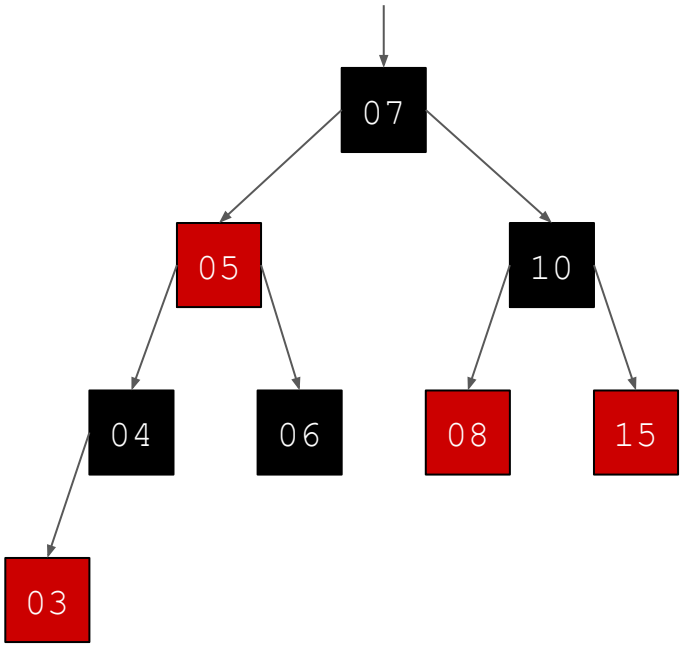
- **Vector** is more performant than **LinkedList**
- **HashMap** is more performant than **TreeMap**

Why?

* with some exceptions (BTrees, link-based graphs, and other use-case specific things)

Red-Black Trees

- Balanced BST
  - `O(log(n))` get, insert, delete
- Similar to AVL
- Each node is colored R/B
  - Just one extra bit per node!
  - Maintain three invariants regarding coloring to keep tree balanced
- HW5: Implement RBTree to act as a KVStore.
- Really nice ownership structure!



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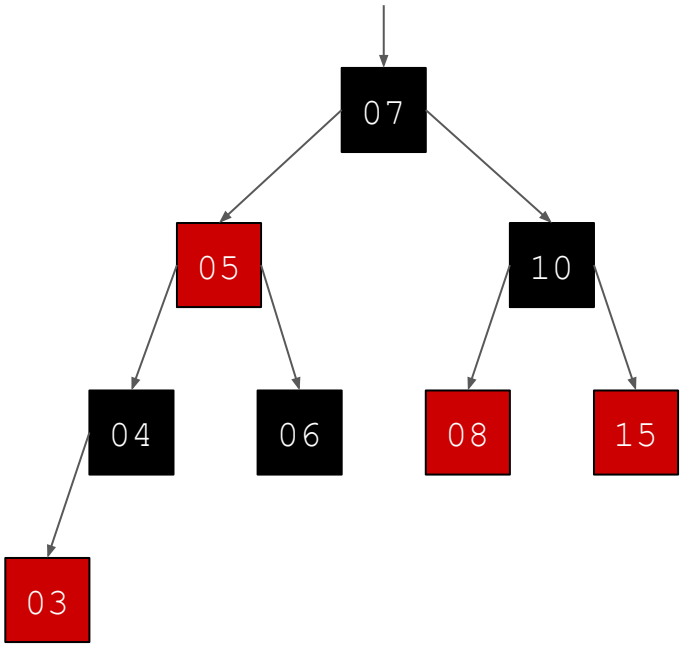

How do we get a value from this tree?

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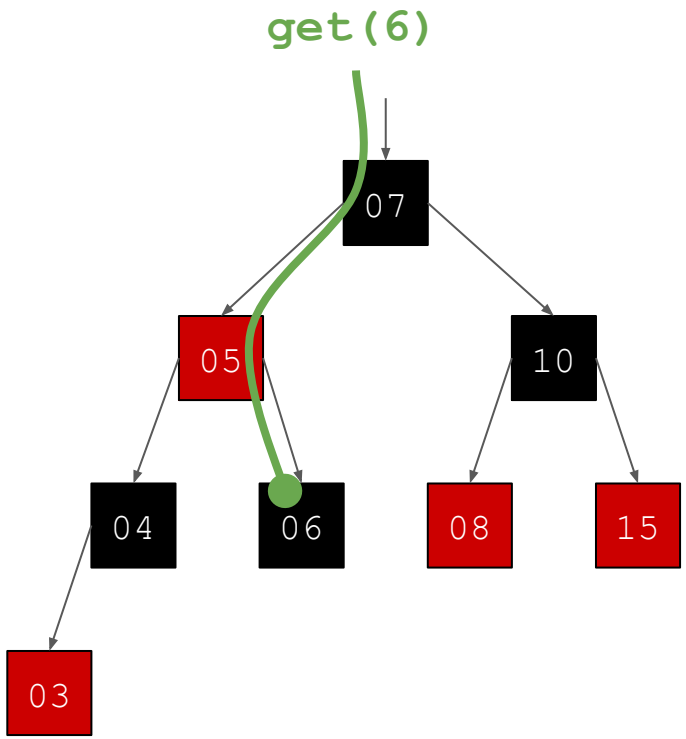
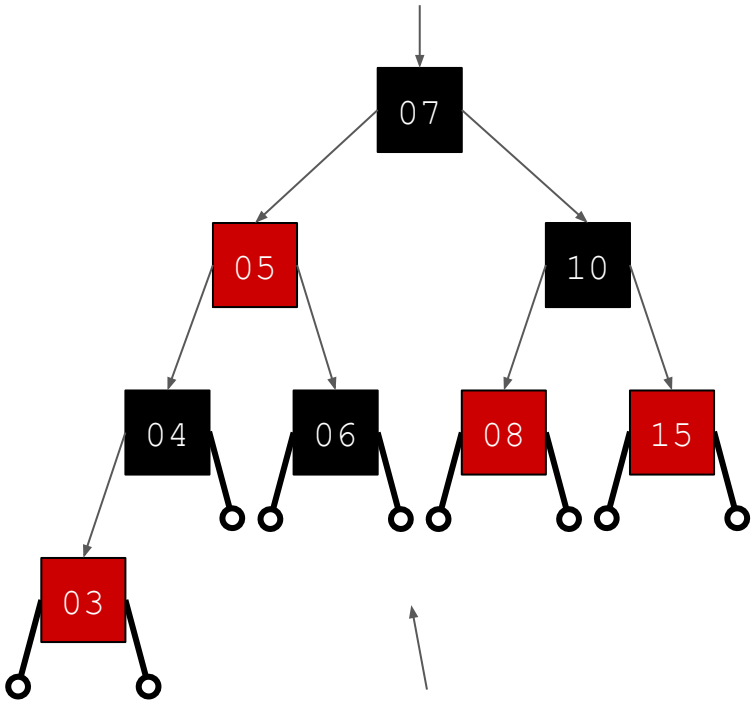


How do we get a value from this tree? *The same way as a regular BST!*

Red-Black Trees: *Insert*

The three invariants that balance the tree:

- **Every node is either red or black. The root node is black. Null nodes are black.**



Let's draw the null nodes too!

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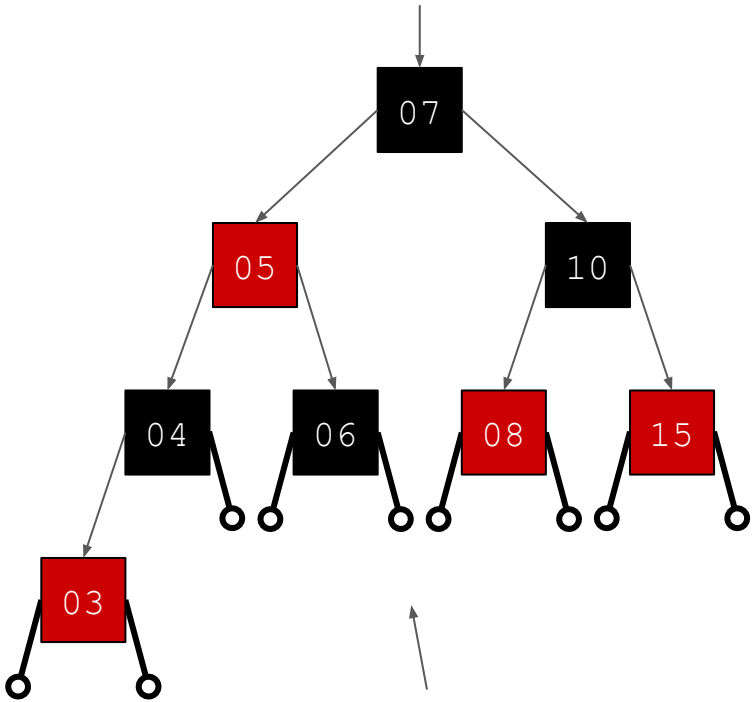

Let's draw the null nodes too!

Generally, fixed by setting the root node to black at the end of `insert`

Red-Black Trees: *Insert*

The three invariants that balance the tree:

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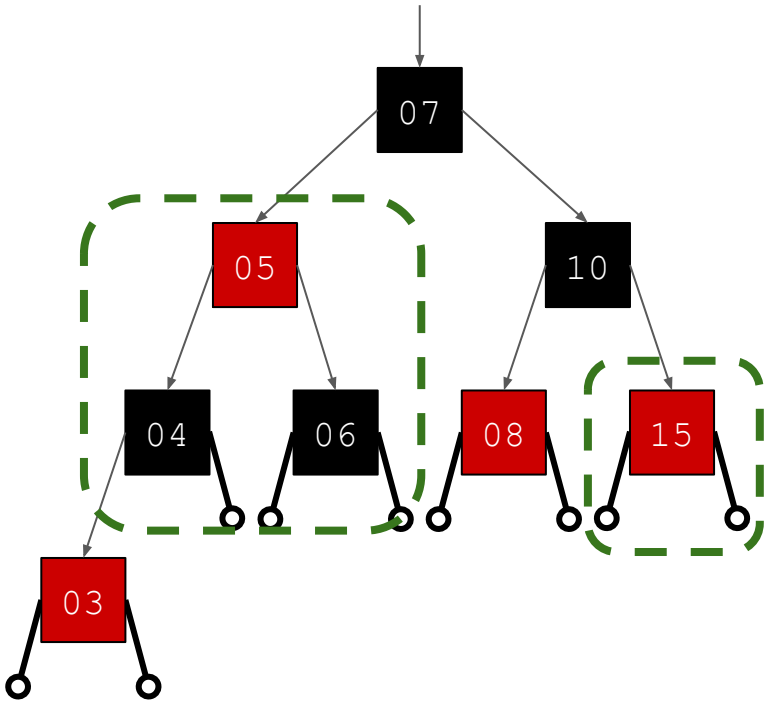

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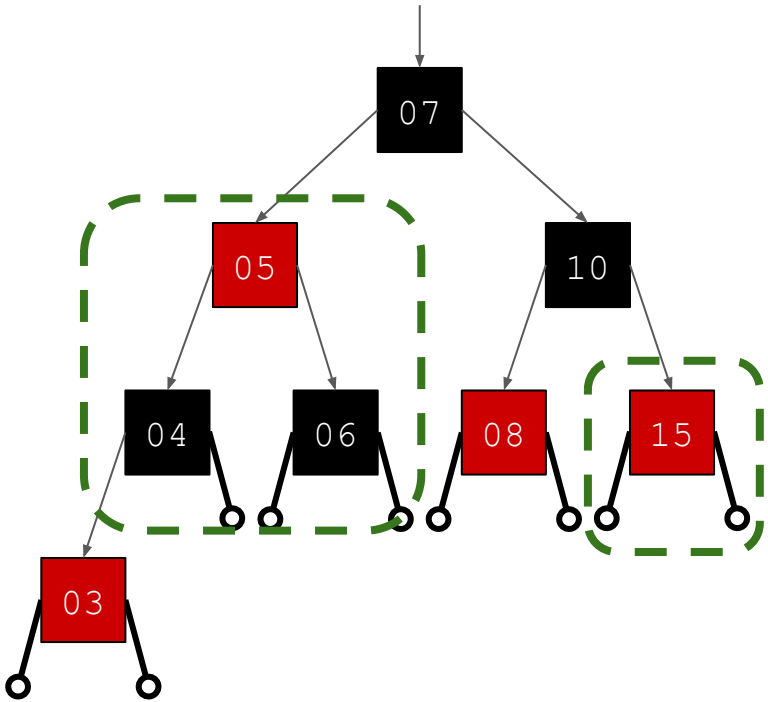

This imposes a rule on how R/B nodes are distributed across levels

Red-Black Trees: *Insert*

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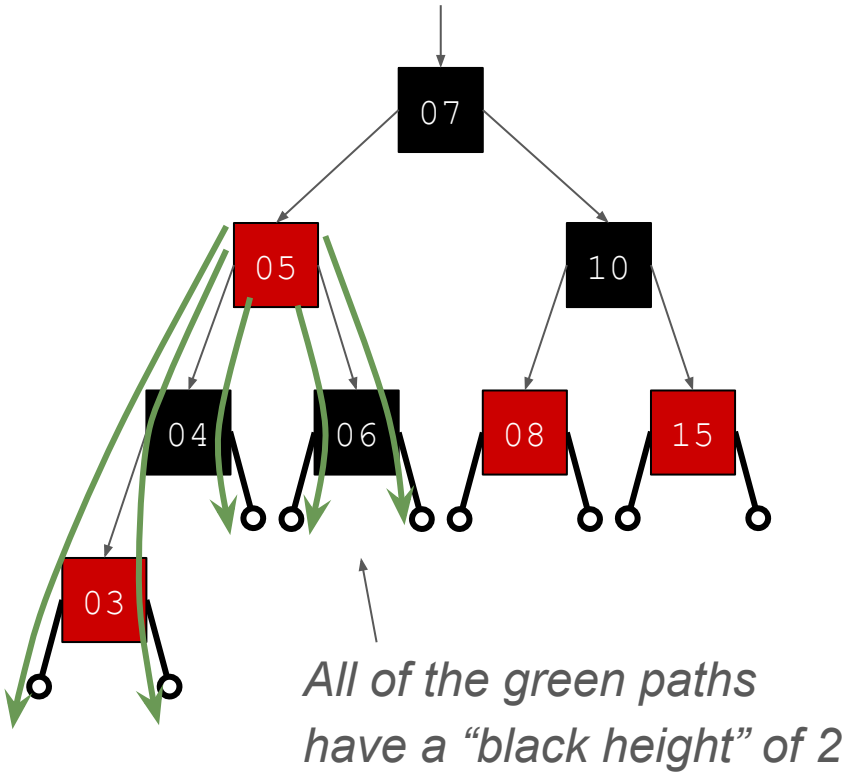

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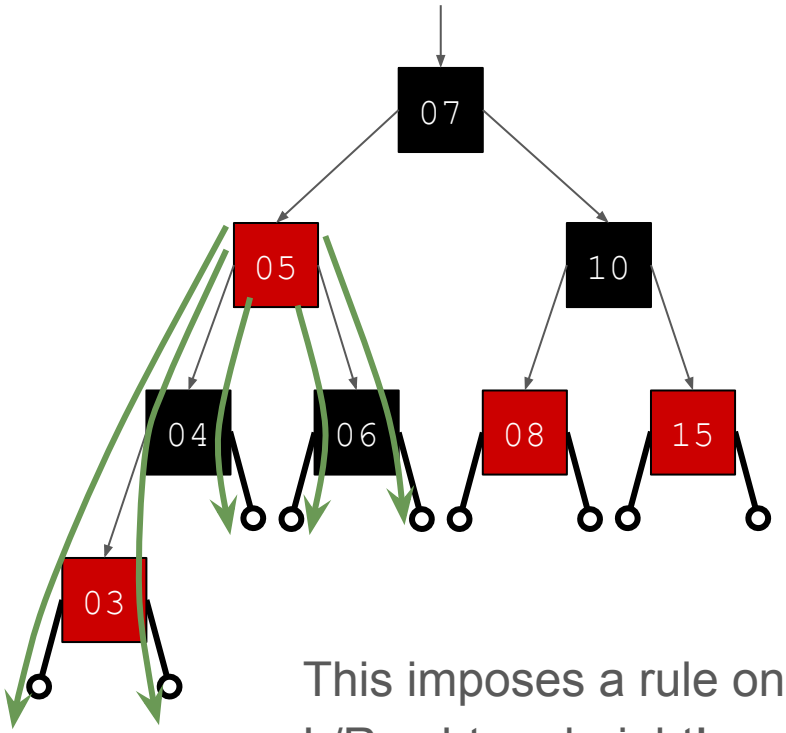
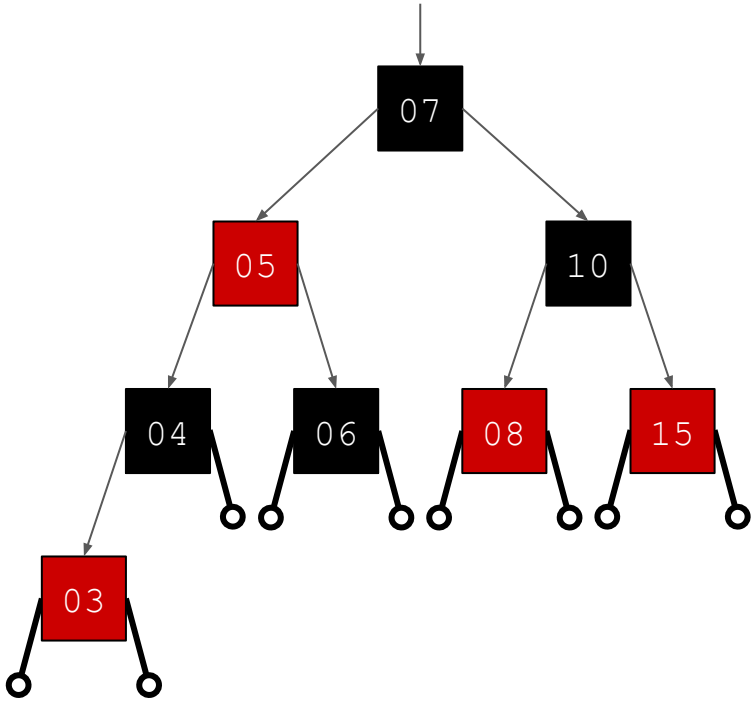


This imposes a rule on L/R subtree height!

Red-Black Trees: *Insert*



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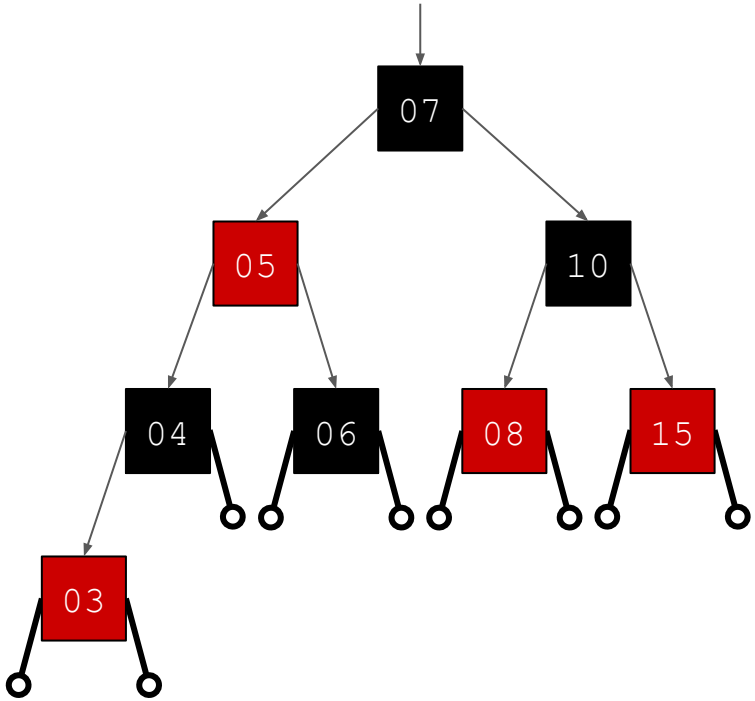

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Let's perform `insert(20)`!

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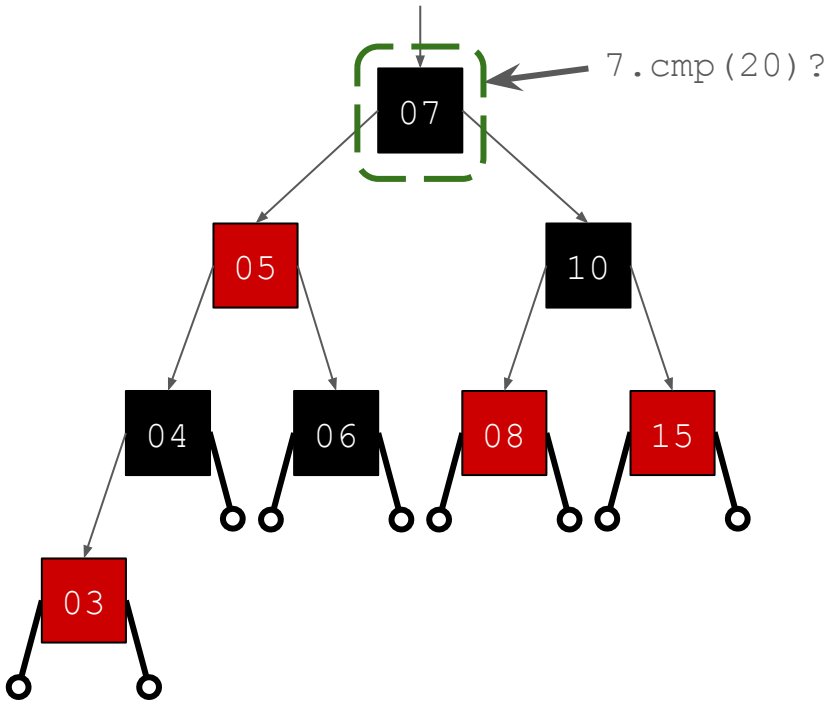

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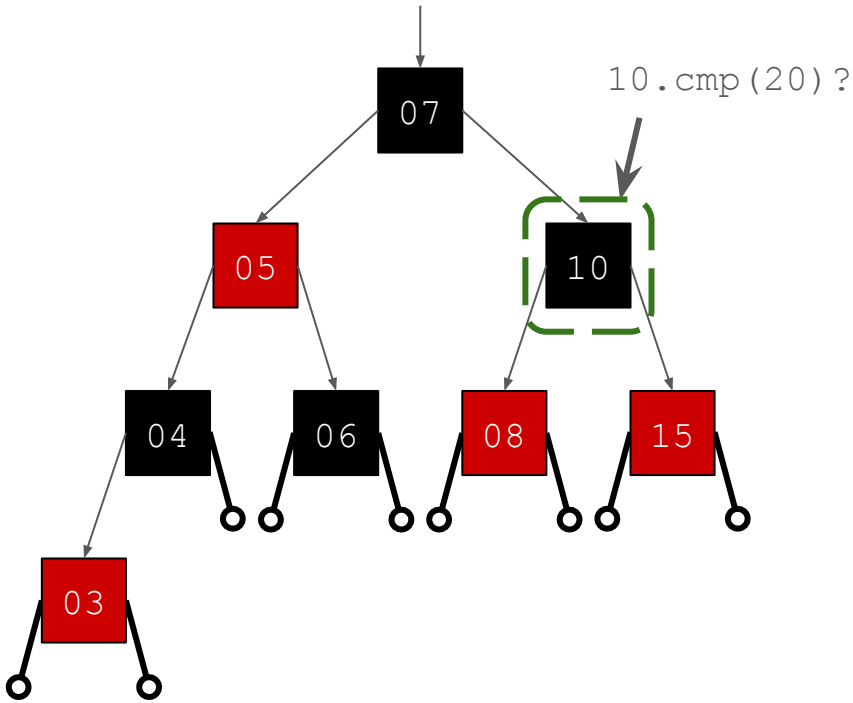

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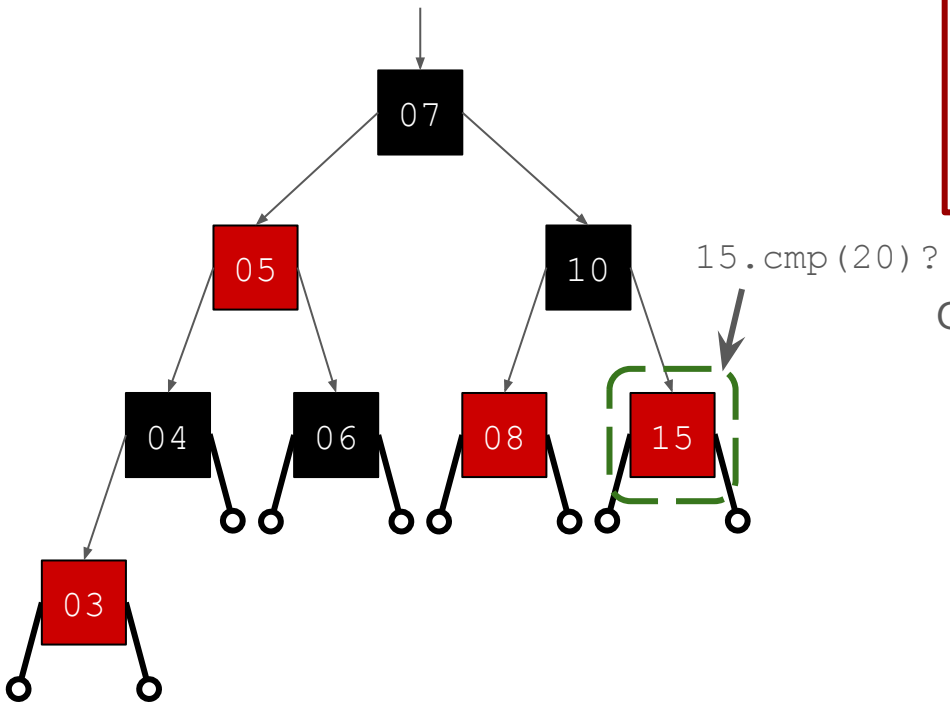

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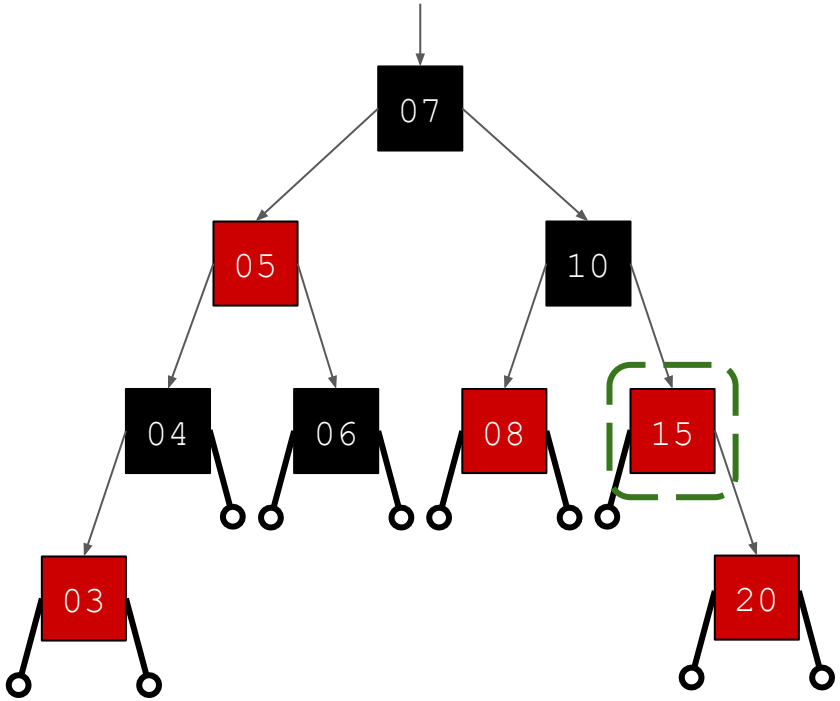

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Newly inserted nodes are *always* red!

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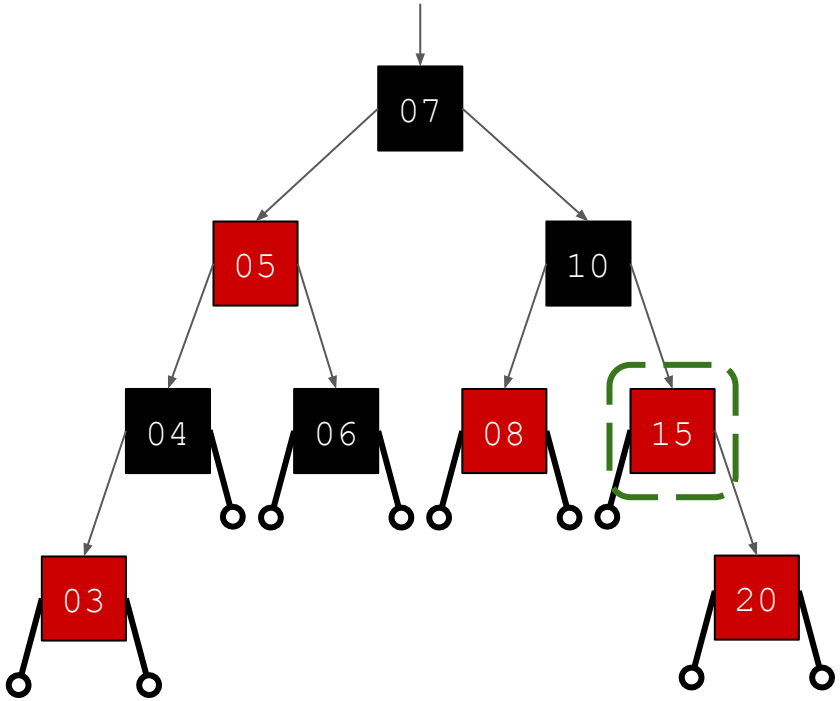

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Are there any violations in the current frame?

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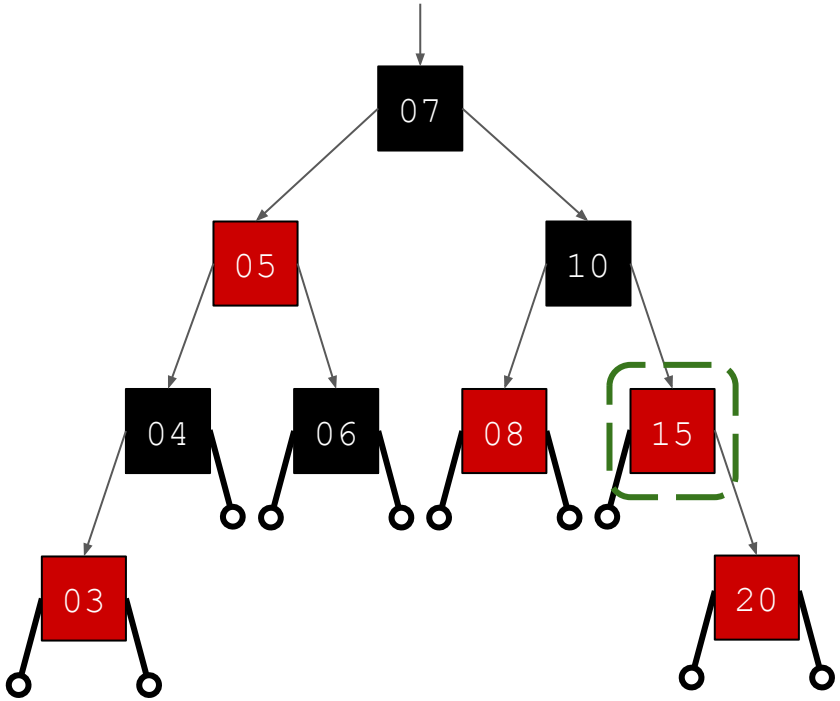

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Are there any violations in the current frame?
Violation of inv. 2!

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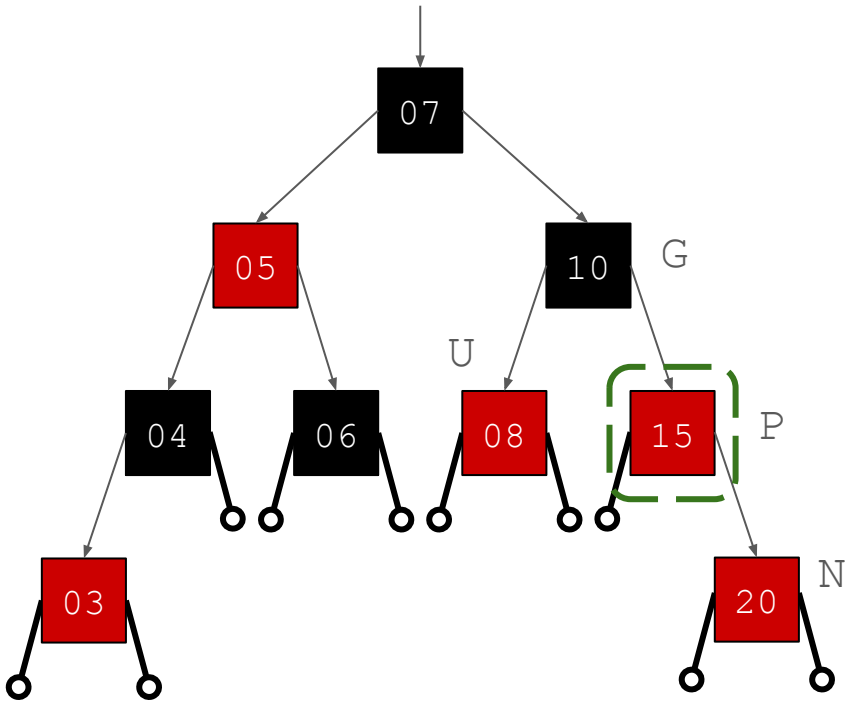

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Let's put some labels on each relevant node to give us some language to talk about the violation

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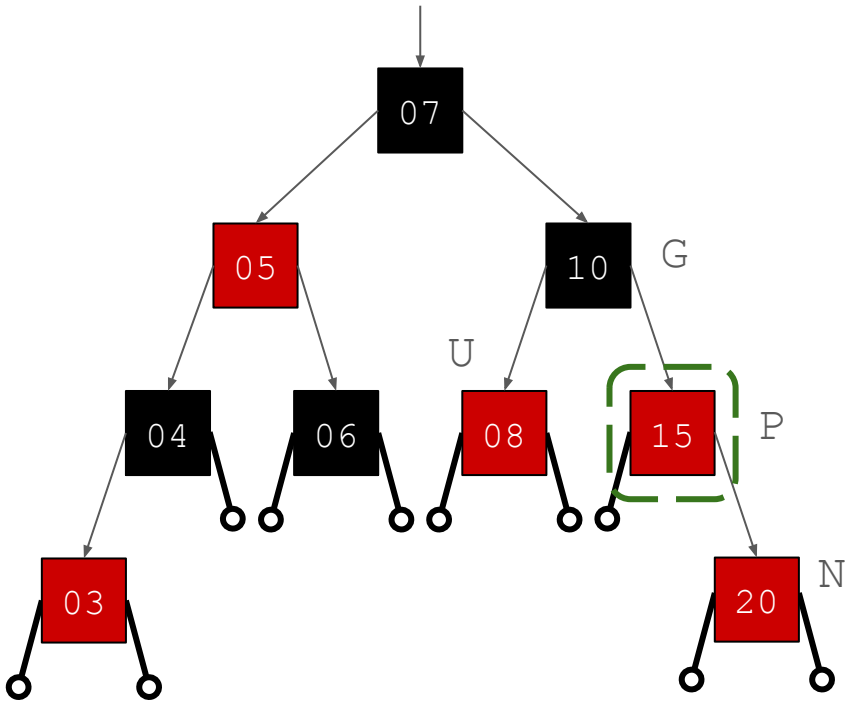

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Because U is red, we can “push down” G’s blackness to fix inv. 2!

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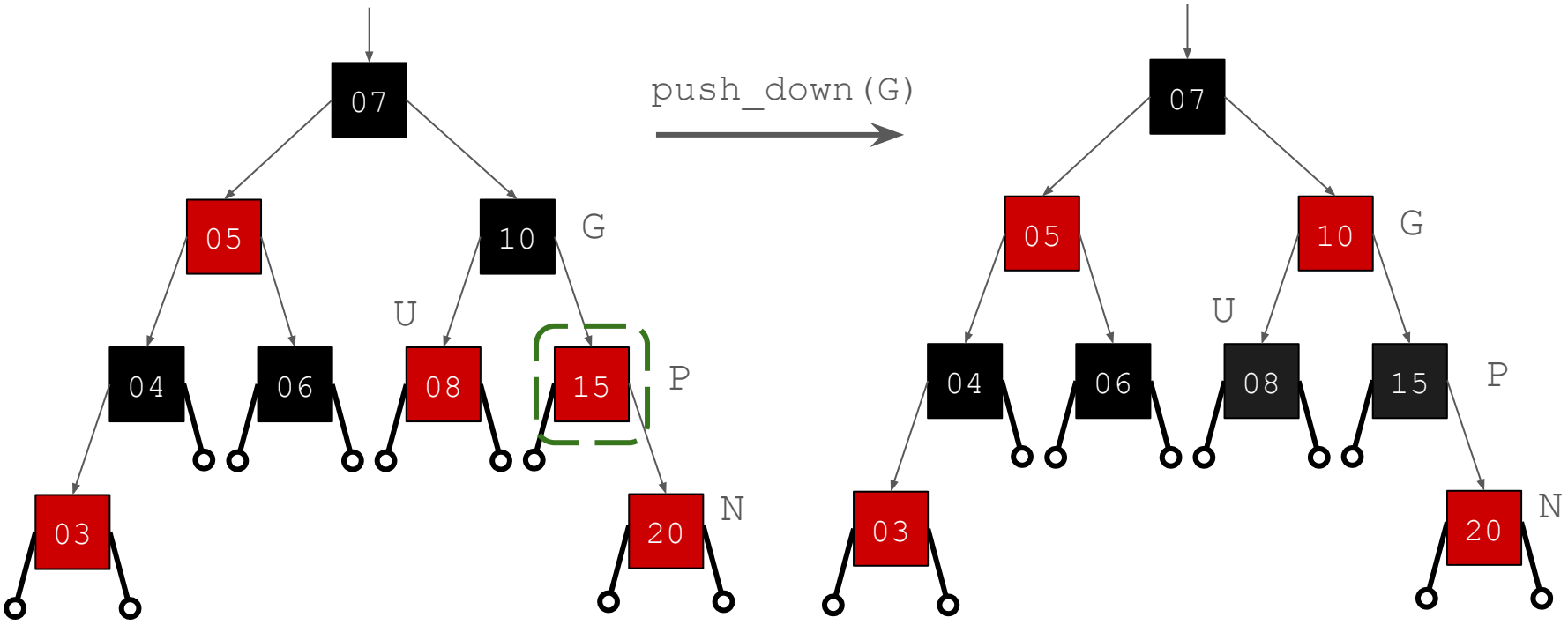

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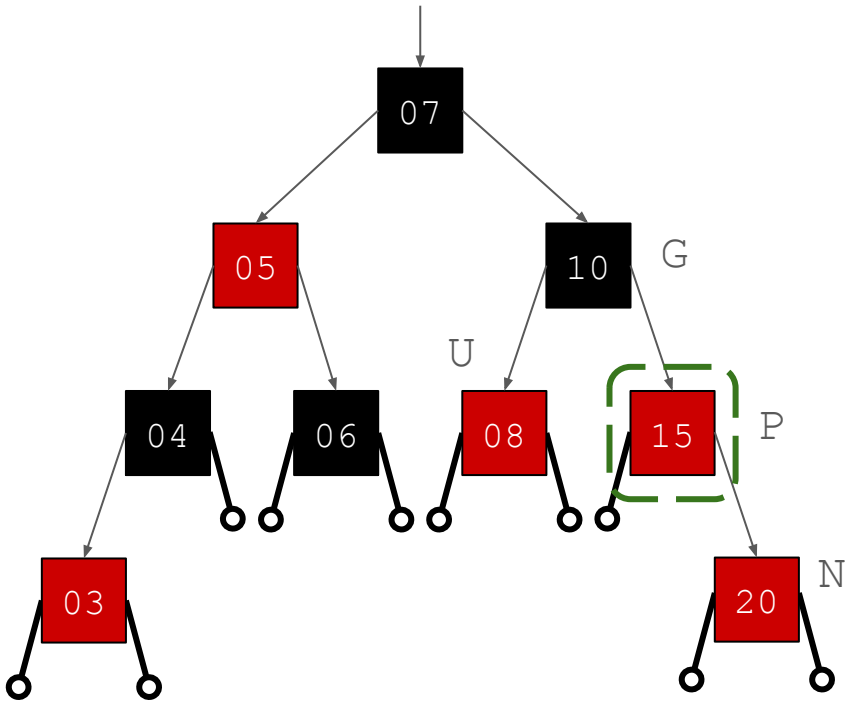

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Wait, to do that, we need information about U! That's outside of the current frame…

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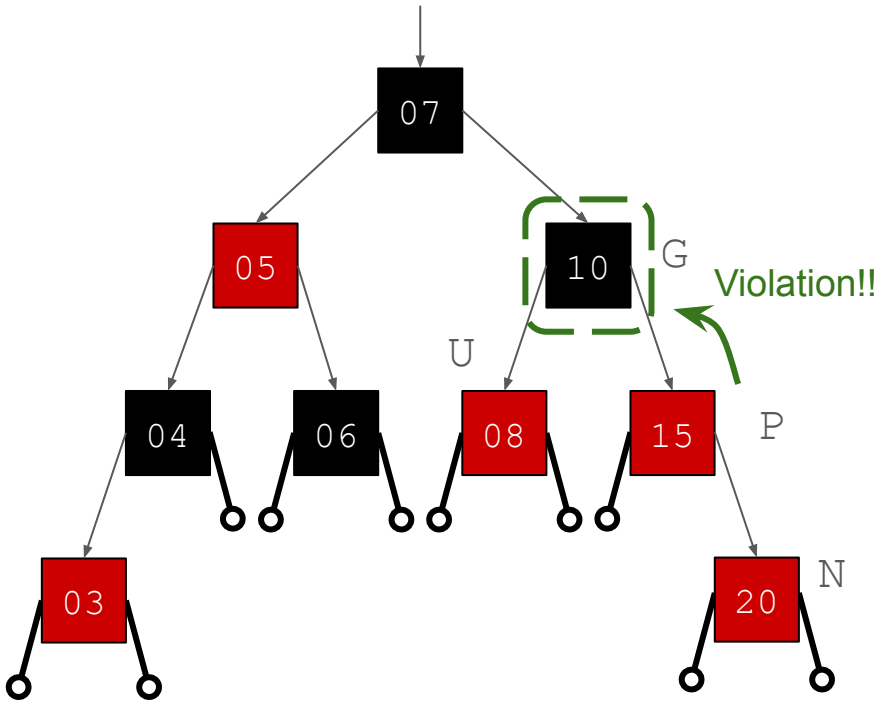

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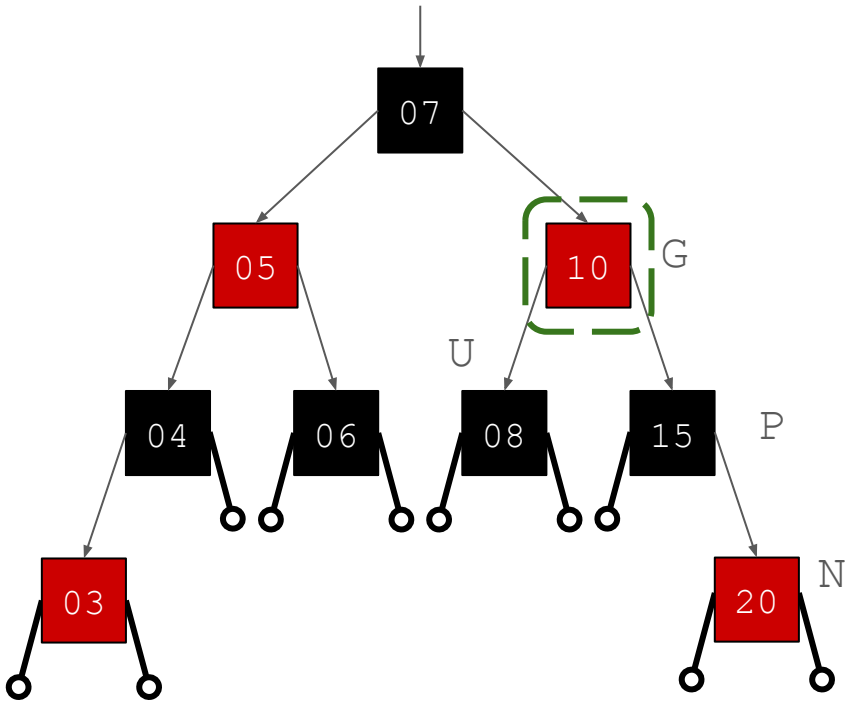
Wait, to do that, we need information about `U`*!*
That's outside of the current frame…

- We need to relay that a violation occurred to `G`...

Red-Black Trees: *Insert*



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`G` finds out that `U` is red, and performs `push_down`

Are we done?

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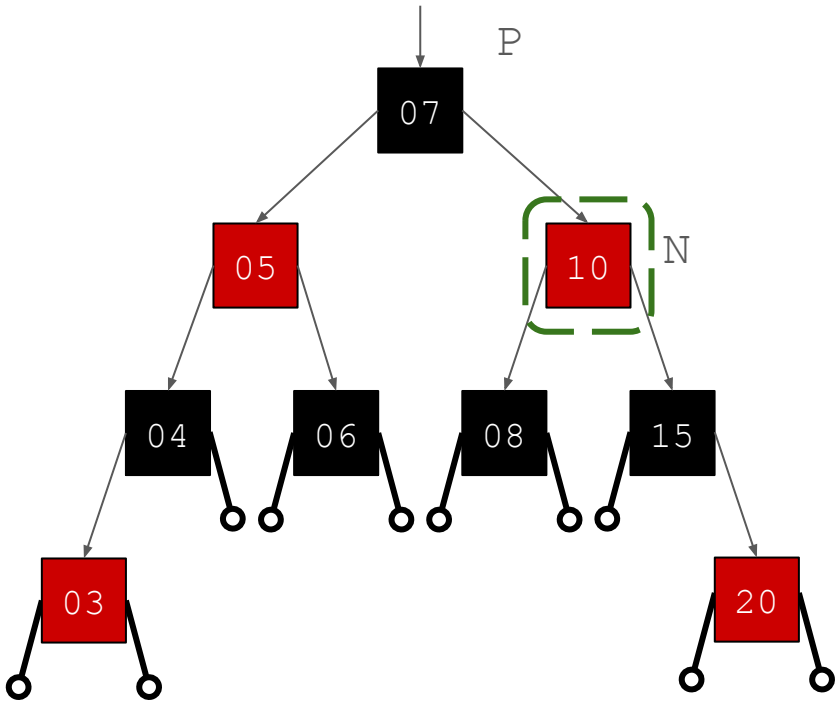

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Changing G to red effectively "inserts" a new red node at G. Depending on G's parent, we could have another violation!

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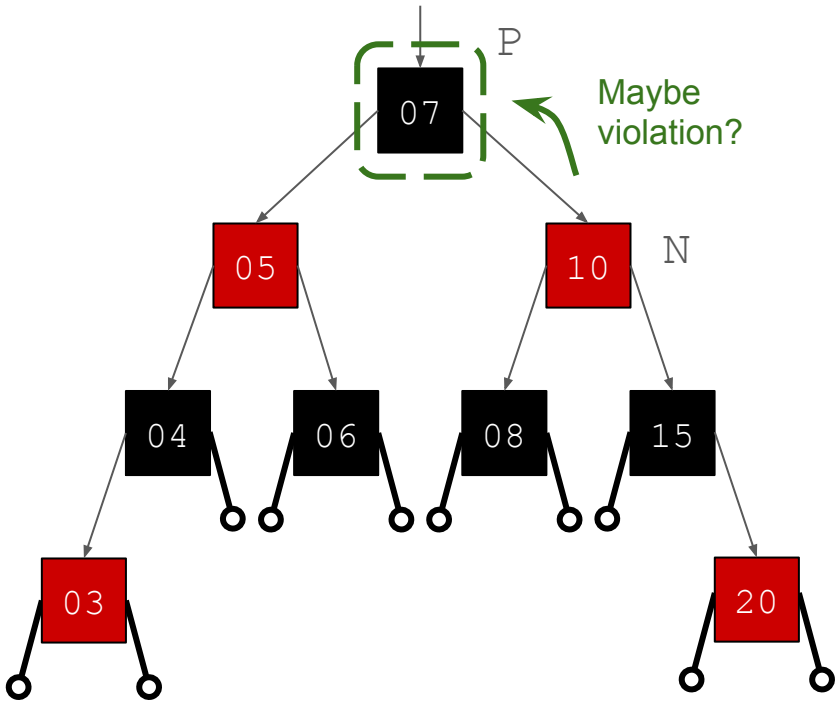

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New P can determine if it's an actual violation, and restart this process

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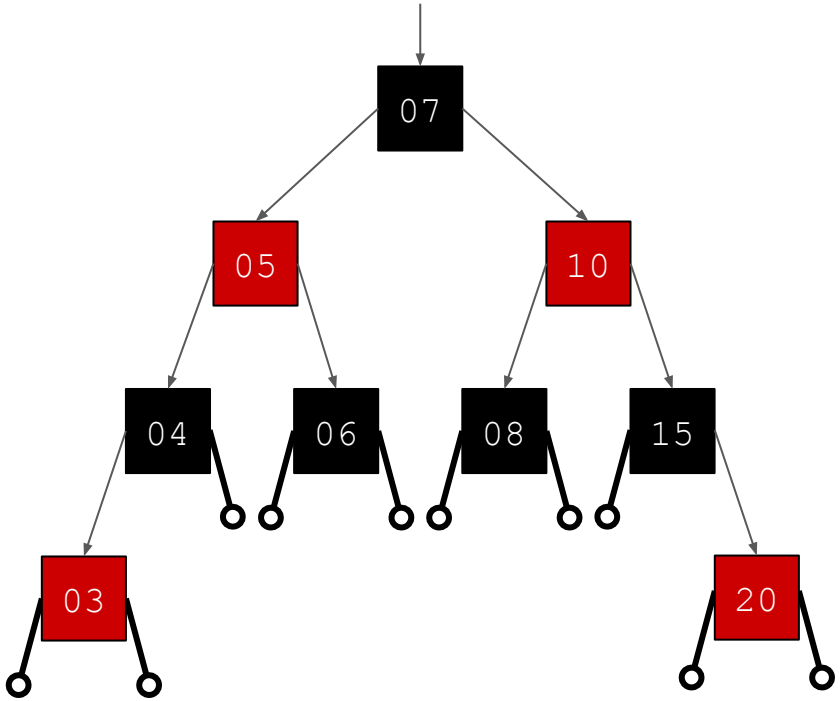

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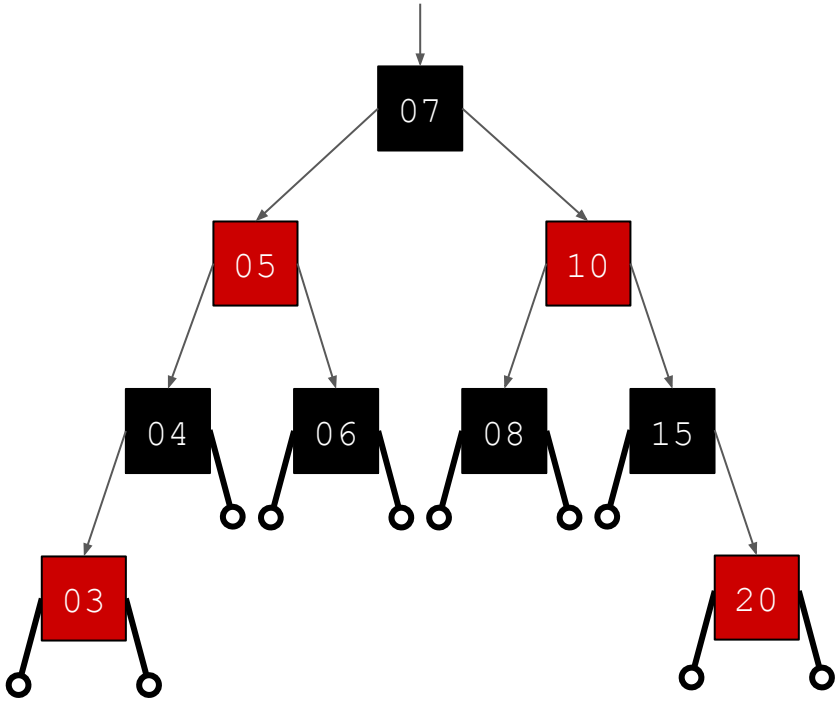
In this case, there is no violation at this level, so we are done!



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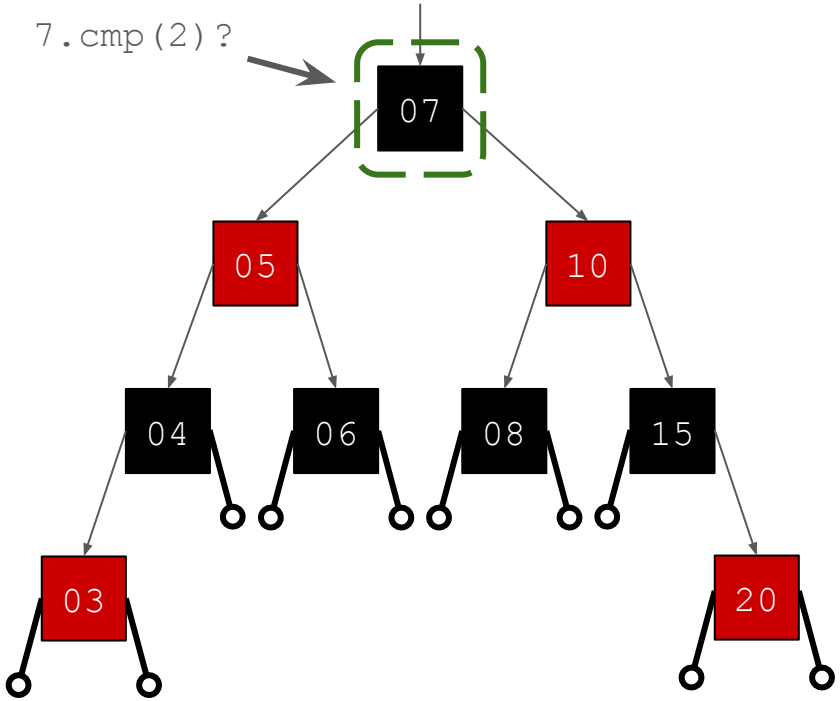

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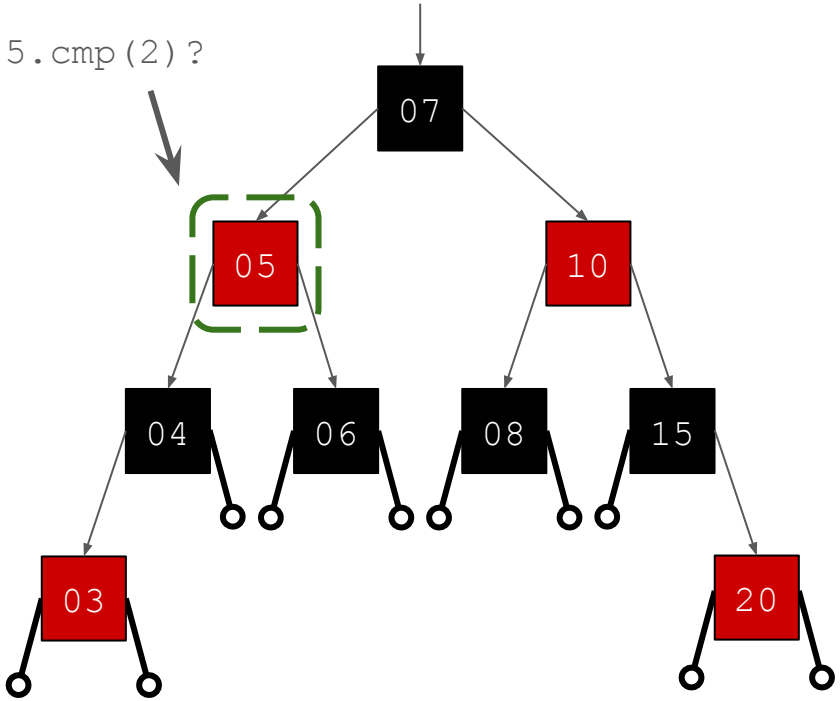

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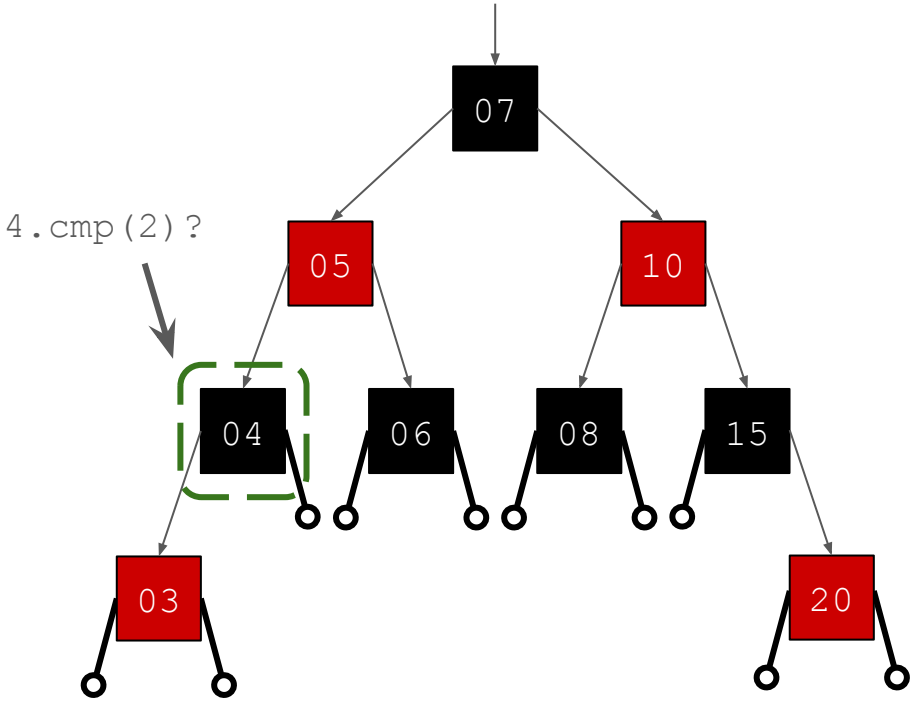
General insertion pseudocode:

1. Perform a standard BST insert (recursively)
2. Detect possible violations while backtracking
3. Fix violations as necessary with rotations and recolors

Let's now do `insert(2)`



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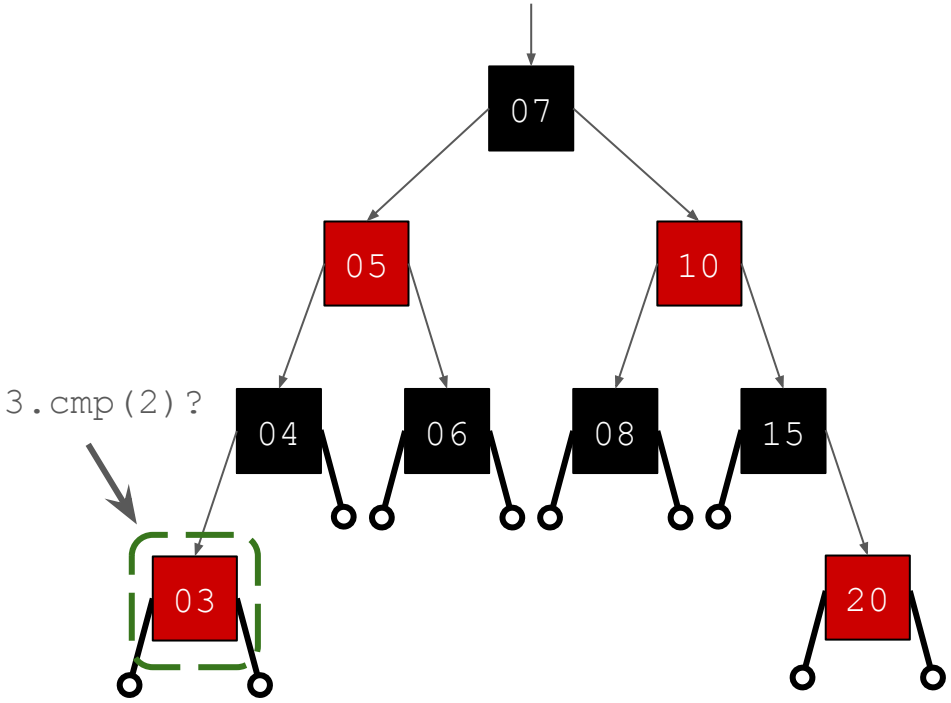

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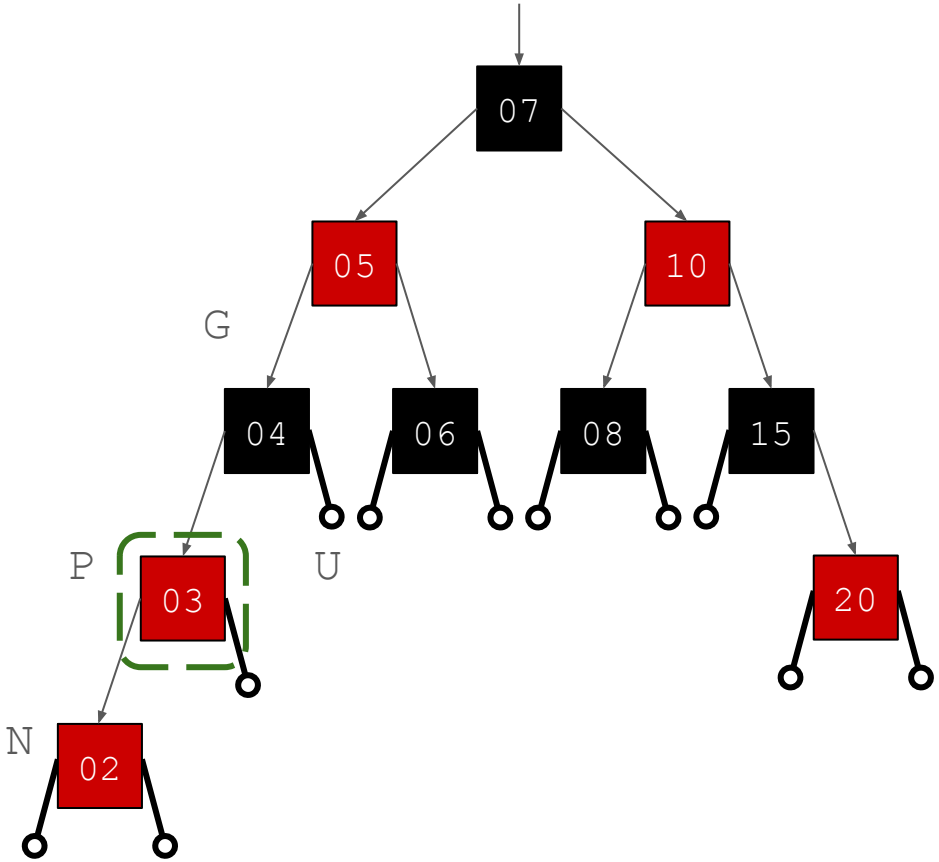

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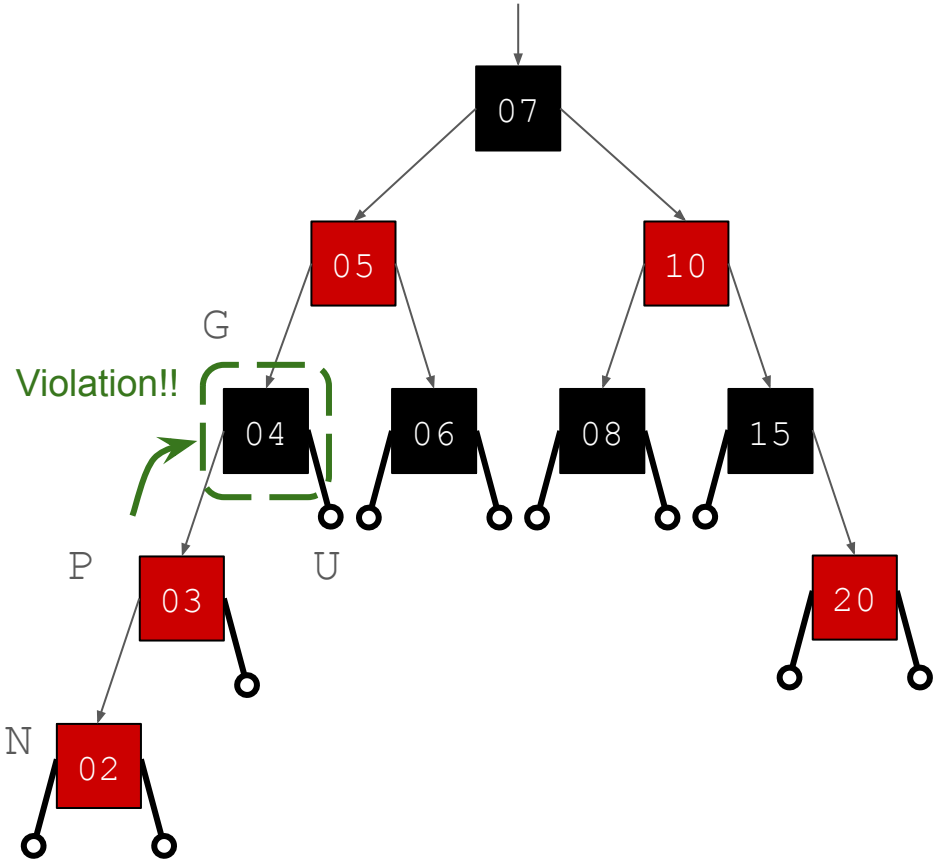
Again, we violate inv. 2…



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General insertion pseudocode:

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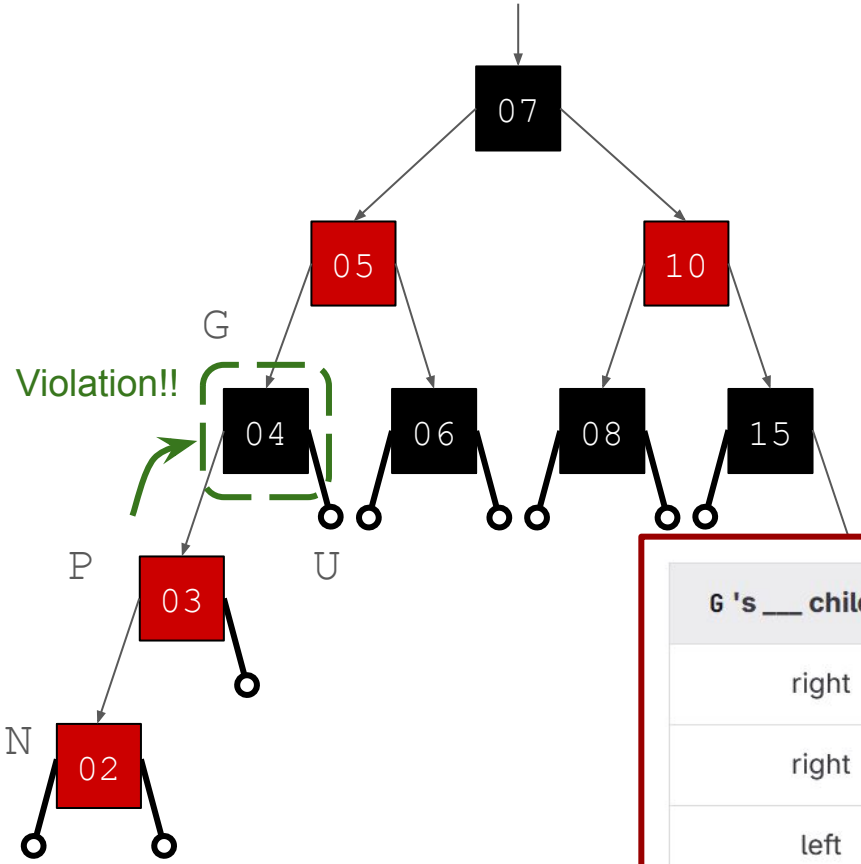
`U` is black – we can't just `push_down`

Fix tree with rotations and recolors



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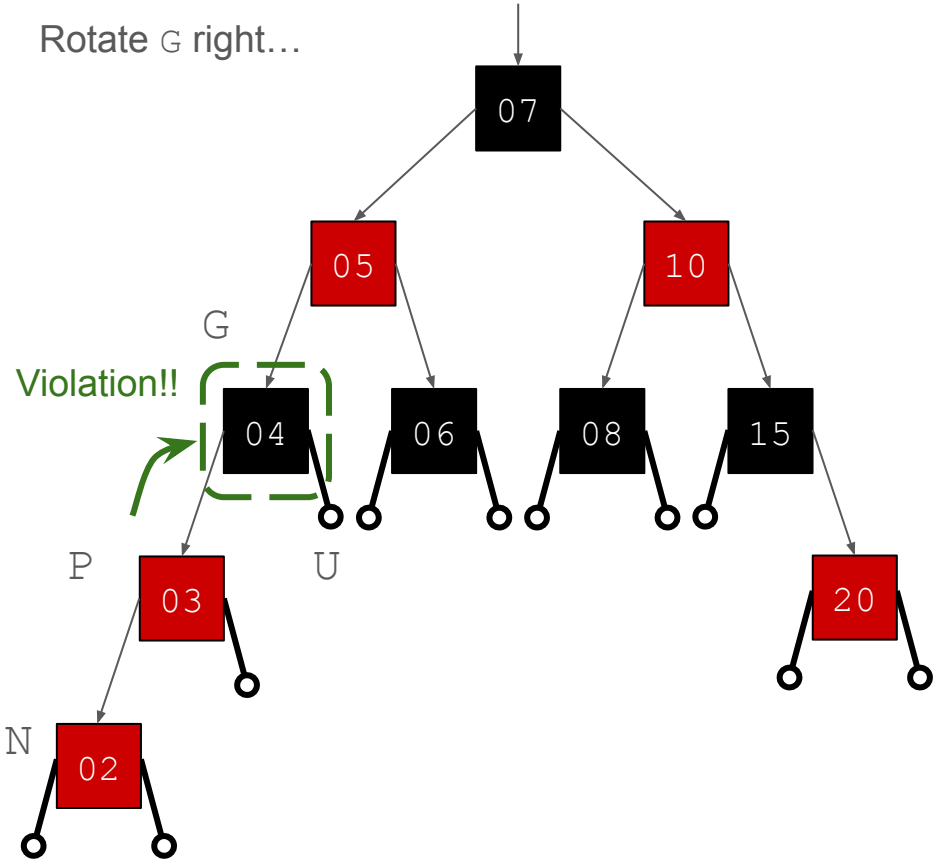
General insertion pseudocode:

1. Perform a standard BST insert (recursively)

G 's ___ child is P	P 's ___ child is N	Rotate	Swap Colors
right	right	G left	G and P
right	left	P right, then G left	G and N
left	right	P left, then G right	G and N
left	left	G right	G and P

Rotate G right…



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General insertion pseudocode:

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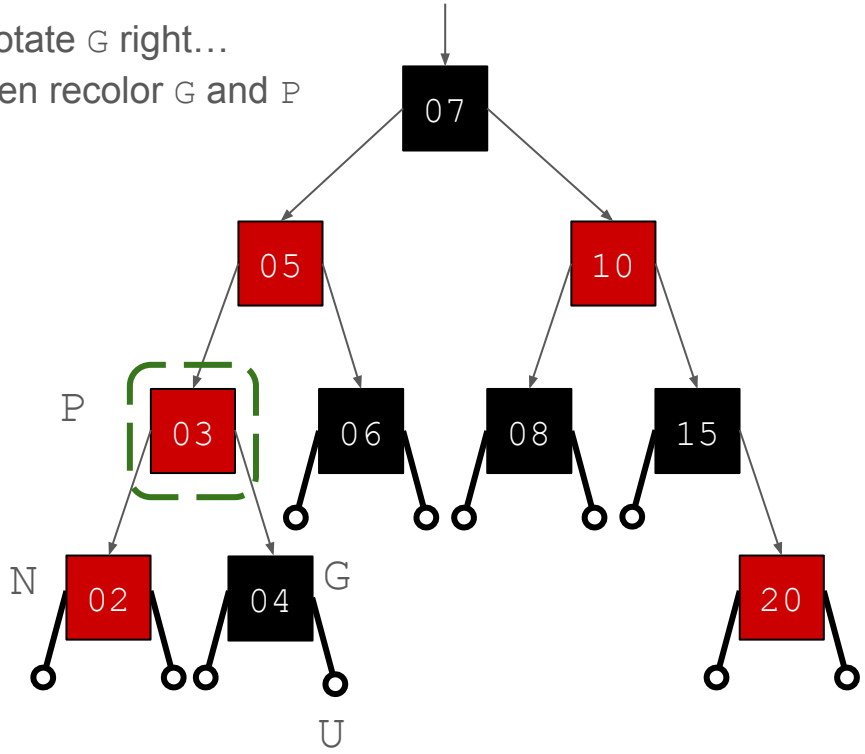
2. Detect possible violations while backtracking
3. Fix violations as necessary with rotations and recolors

Because of the L/R relationships of G, P, and N, we need to rotate G right, then swap G and P's color

Rotate G right…
then recolor G and P



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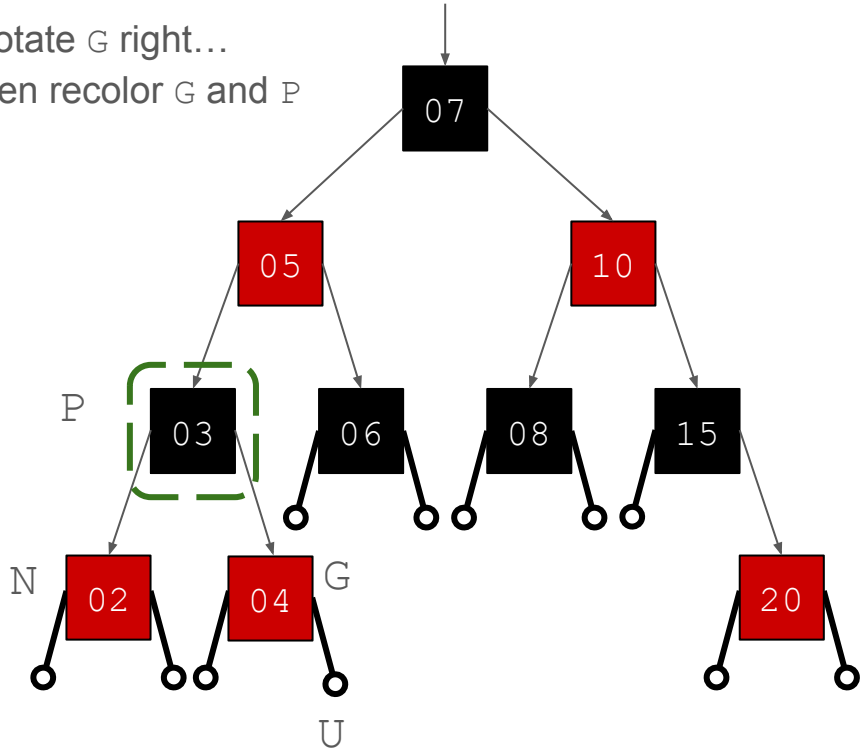

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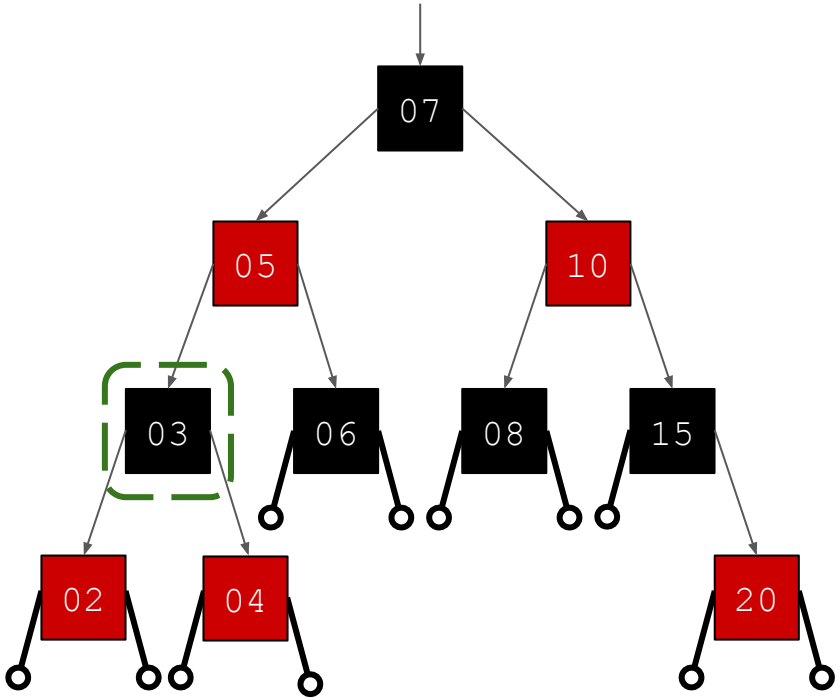

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General insertion pseudocode:

1. Perform a standard BST insert (recursively)
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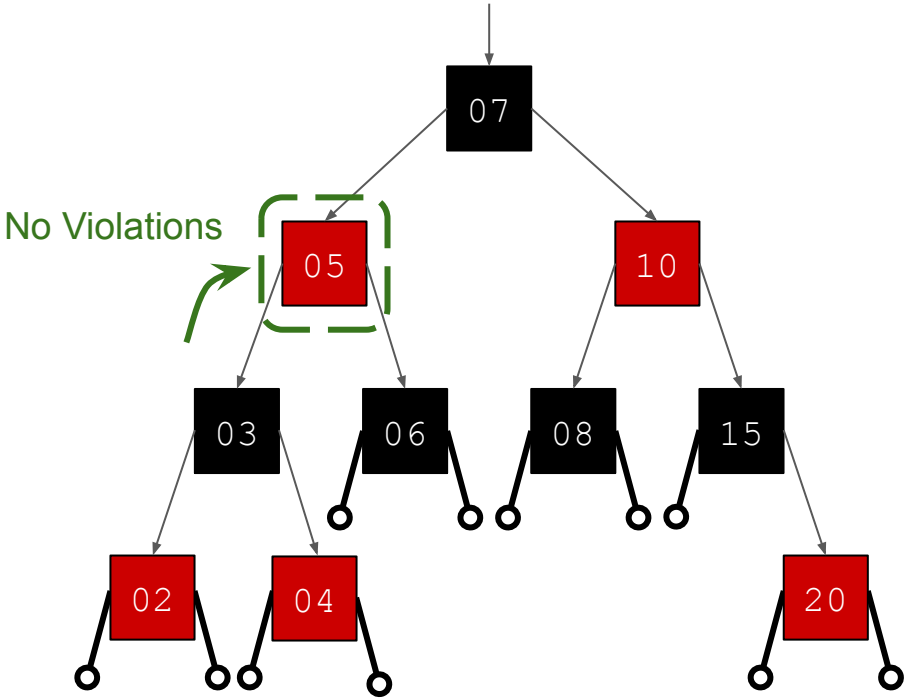
Current frame ends up with a black node, so there are no more violations!



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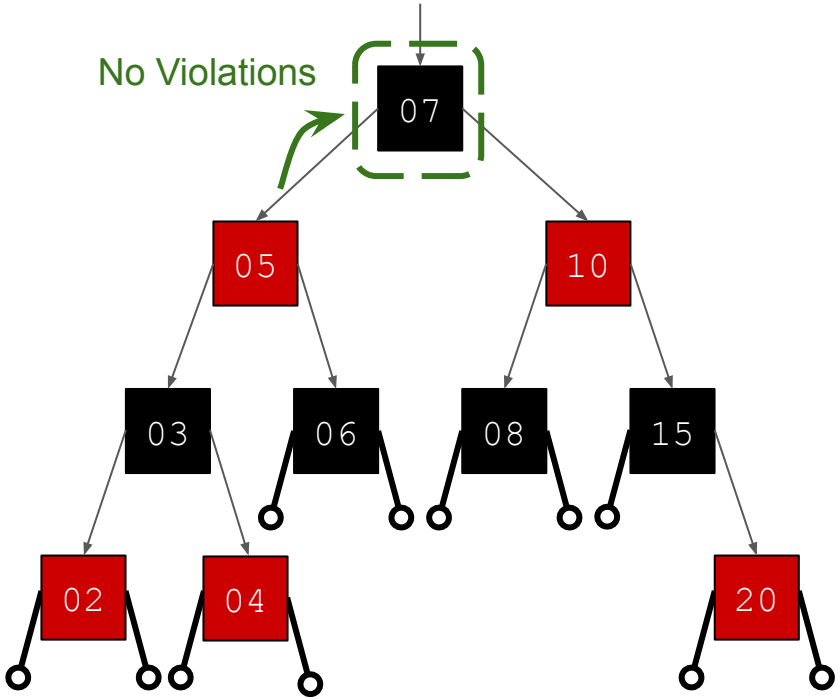

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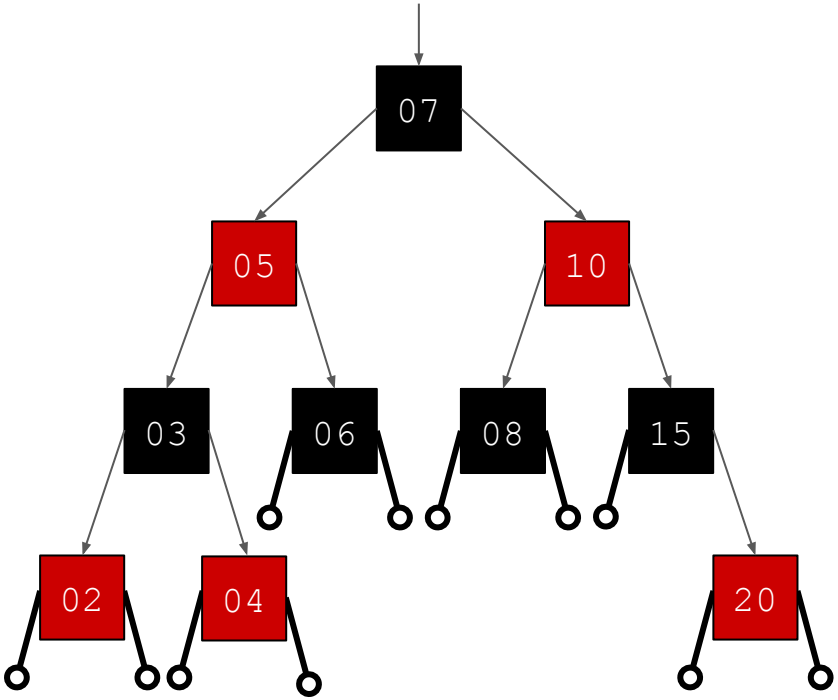
Red-Black Trees: *Insert*



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General insertion pseudocode:

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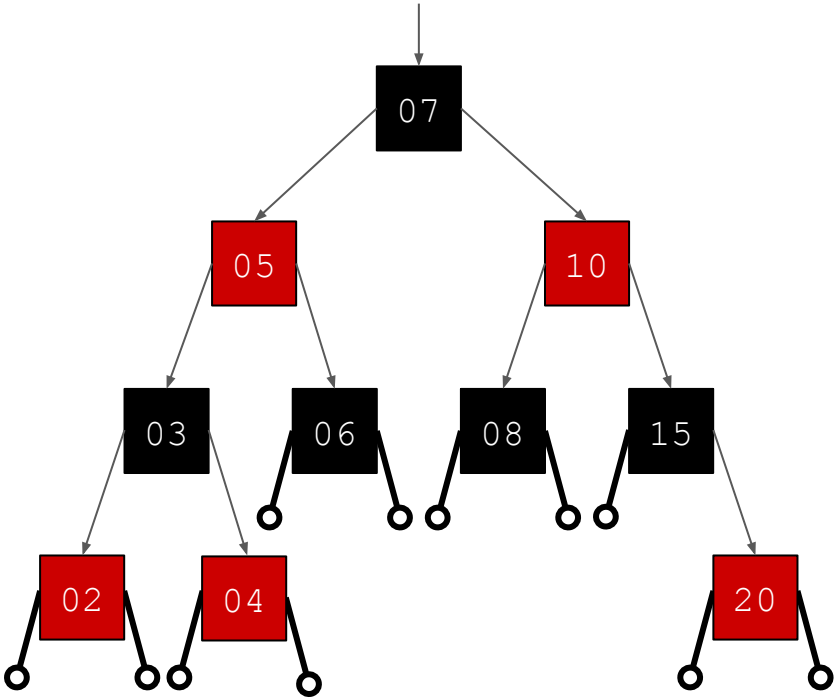
Our tree is balanced again!

Red-Black Trees: *Insert*



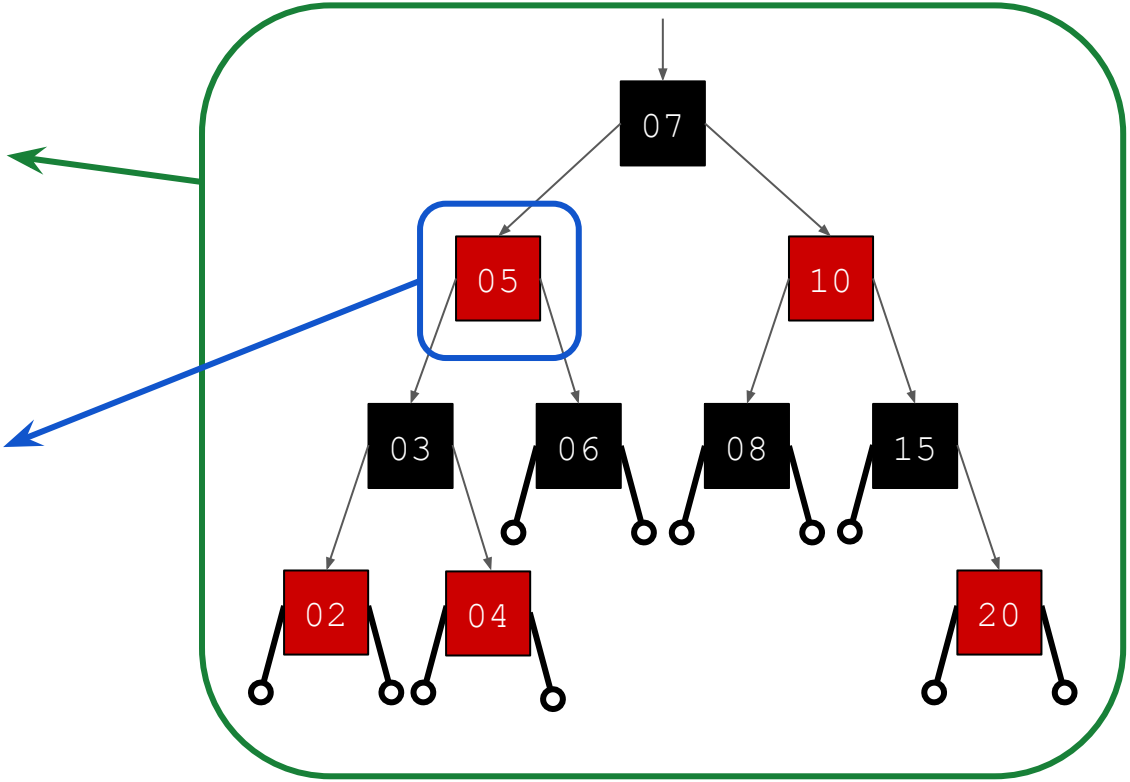
More things to keep in mind:

- Carefully consider what the return type is from any recursive `insert()` call
  - Encode all information necessary to be able to decide how to fix the tree
- At the end of insertion, if the root node changes to red, change it to black
- If the insertion is for a key that already exists, `RBTree.insert()` should replace it, and return the old value

Red-Black Trees: *Rust-Specific Stuff*

```rust
pub struct RBTree<K, V>
where
    K: Ord,
{
    root: Option<Box<Node<K, V>>>,
    size: usize,
}
```

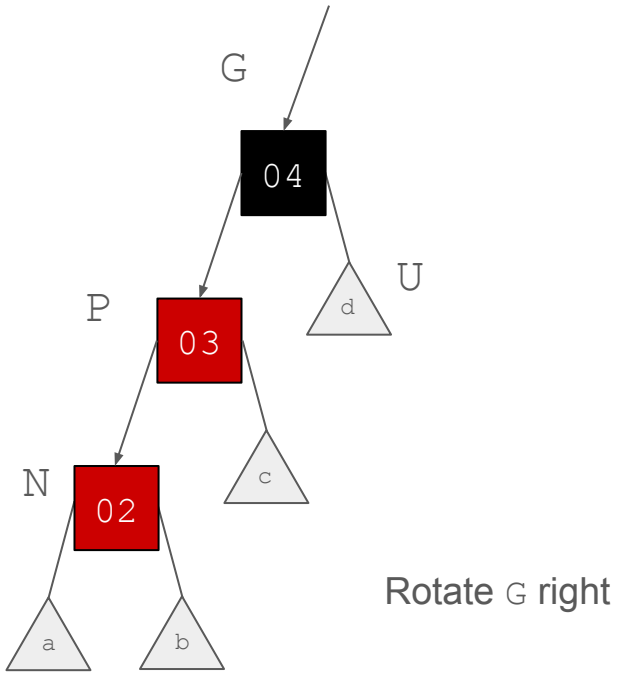
```rust
pub struct Node<K, V>
where
    K: Ord,
{
    pub key: K,
    pub value: V,
    pub color: Color,
    right: Option<Box<Node<K, V>>>,
    left: Option<Box<Node<K, V>>>,
}
```



Red-Black Trees: *Rust-Specific Stuff*

Rotations are harder than usual…

- Remember that you cannot have "partially constructed types"
- Making a temporary variable to perform the pointer swap won't work…



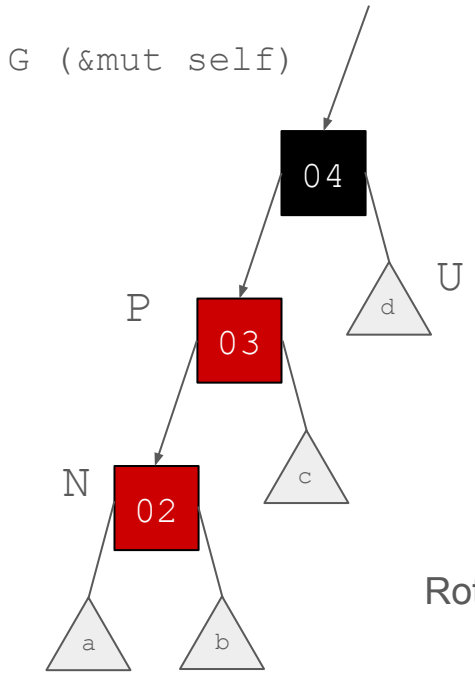
Rotate G right

Drawing subtrees now…

Red-Black Trees: *Rust-Specific Stuff*

Rotations are harder than usual…

1. Does `self.left` exist?
   a. If not, then we can't rotate → no-op
   b. If so, then we can rotate!

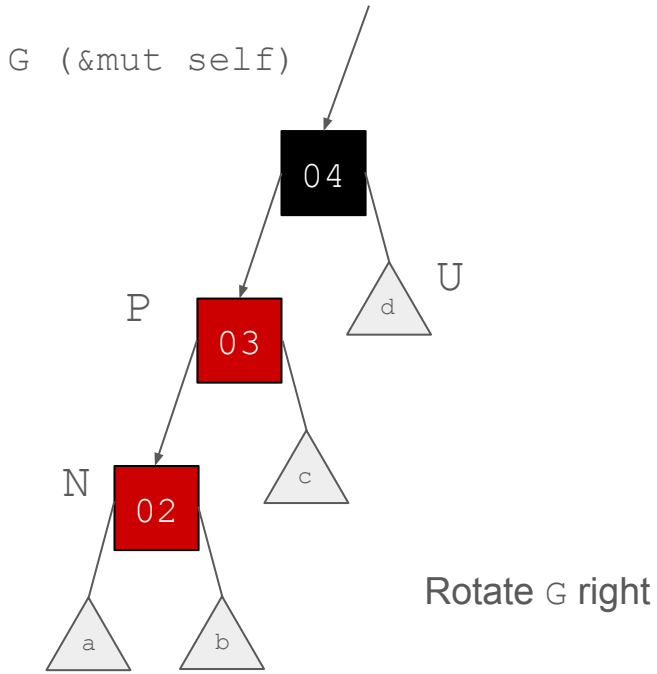


Rotate `G` right

Red-Black Trees: *Rust-Specific Stuff*

Rotations are harder than usual…

1. Does `self.left` exist?
2. Extract `self.left.left` and `self.left.right`
   a. Need to pull the value out of the `Option`, leaving a `None` in their place*



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* README has some helpful functions to do this

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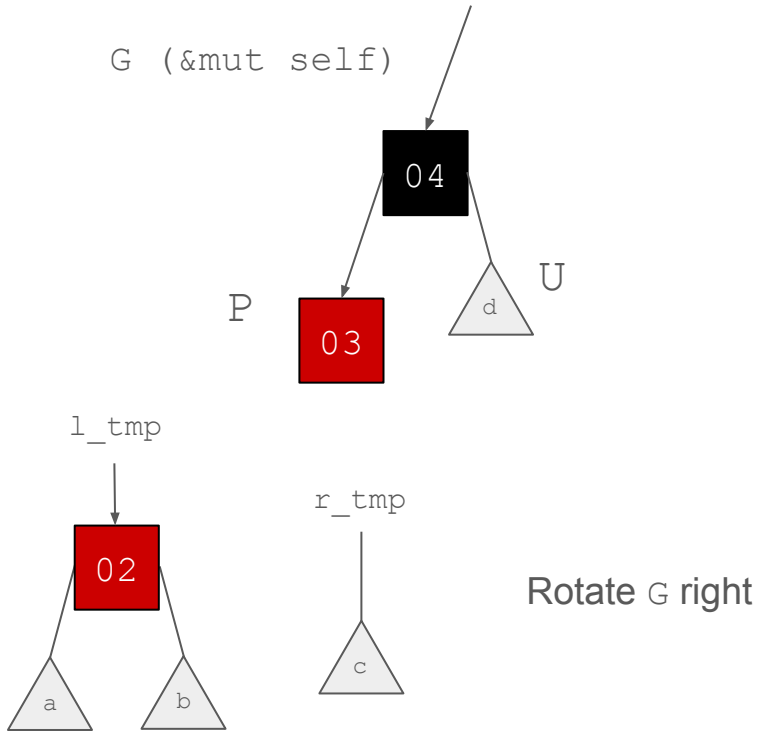

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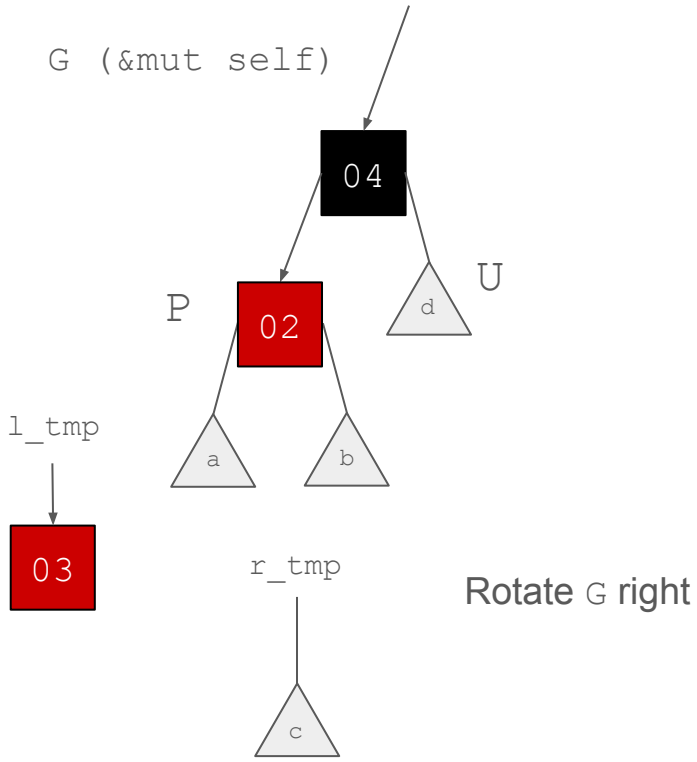
3. Replace `P` contents with `l_tmp`



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Red-Black Trees: *Rust-Specific Stuff*

Rotations are harder than usual…

1. Does `self.left` exist?
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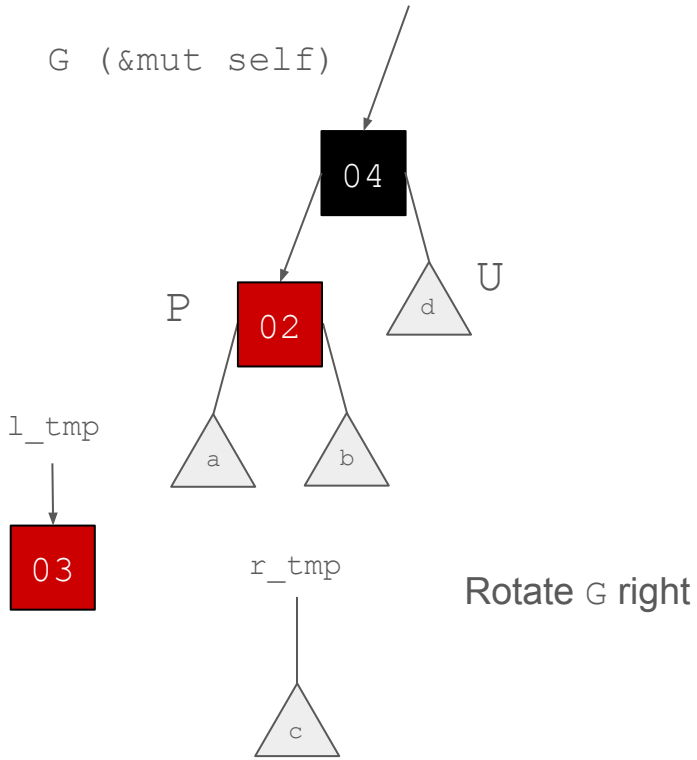
4. Swap contents of `self` and `l_tmp`



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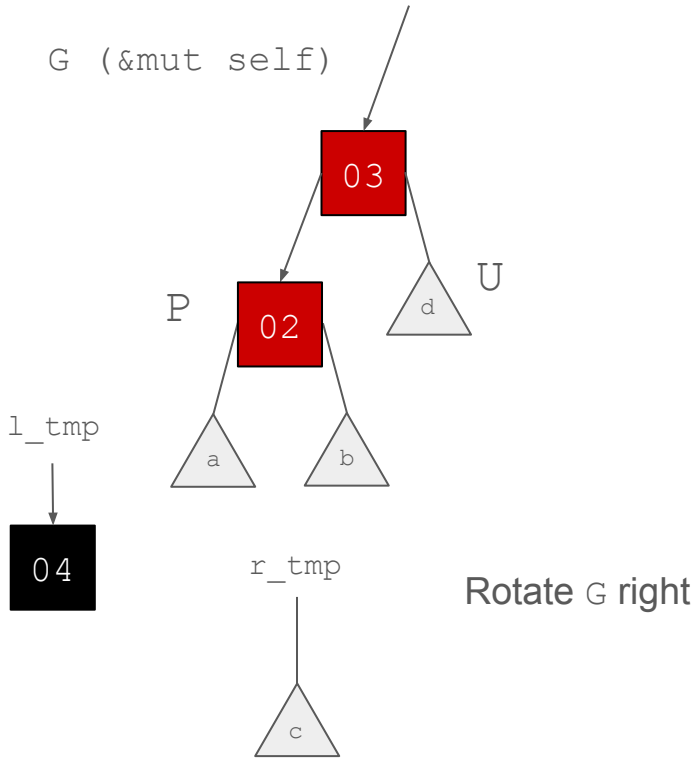

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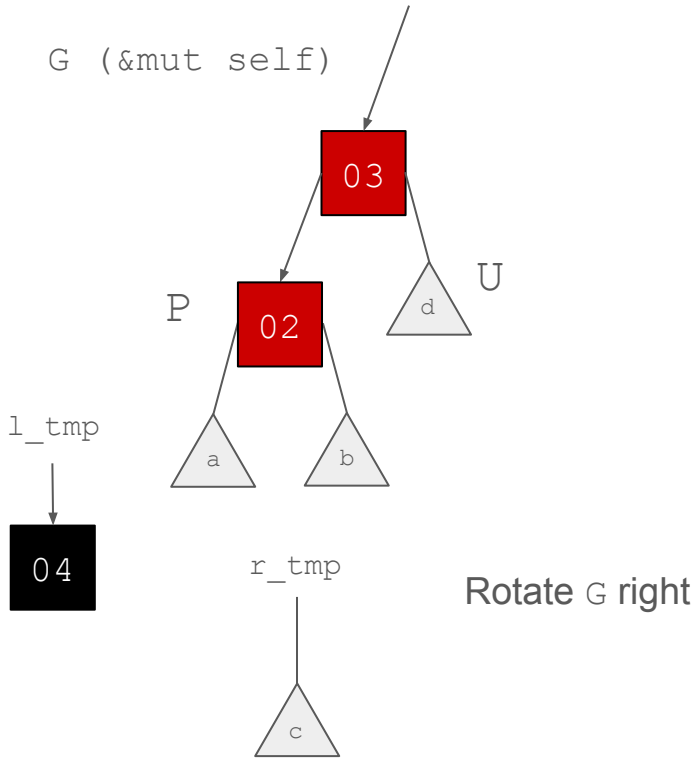
5. Extract `self.right`



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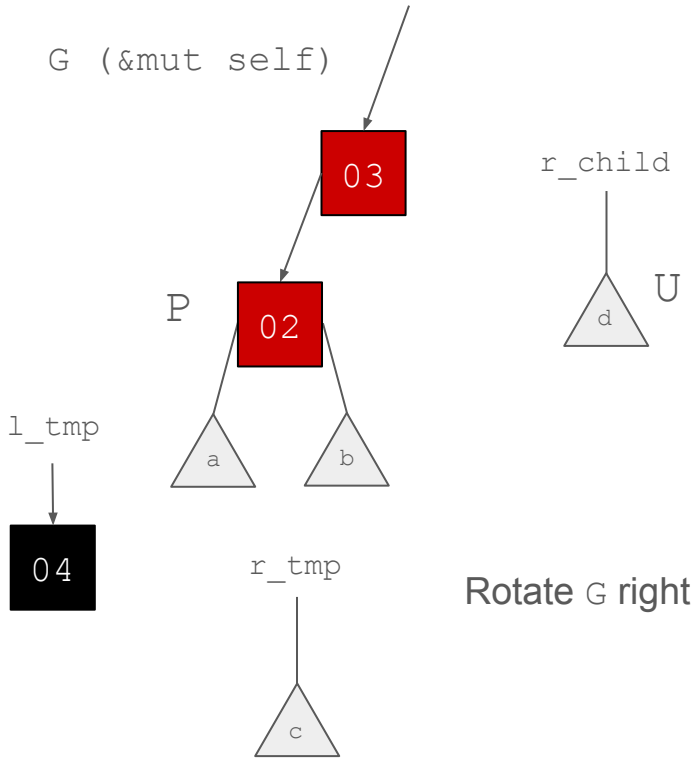

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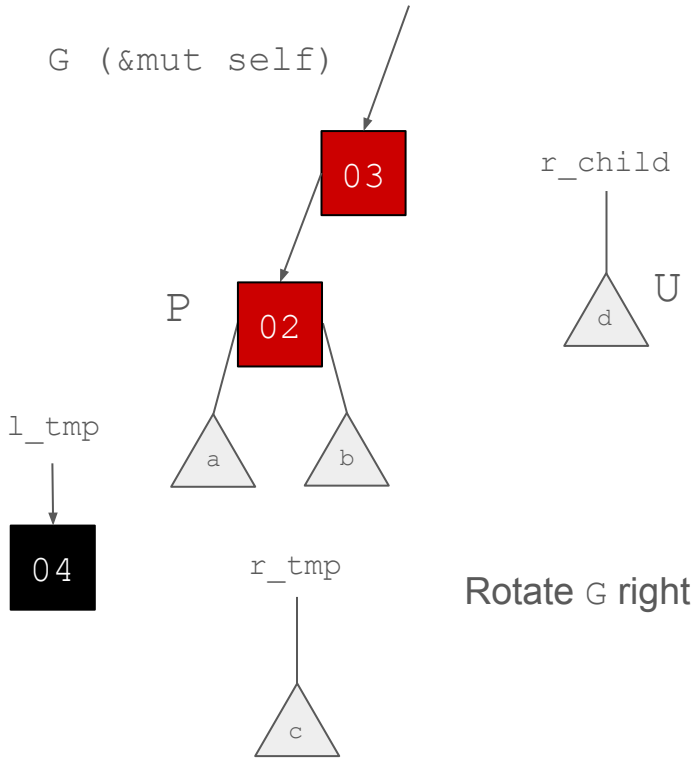
5. Extract `self.right`
6. Construct new right child



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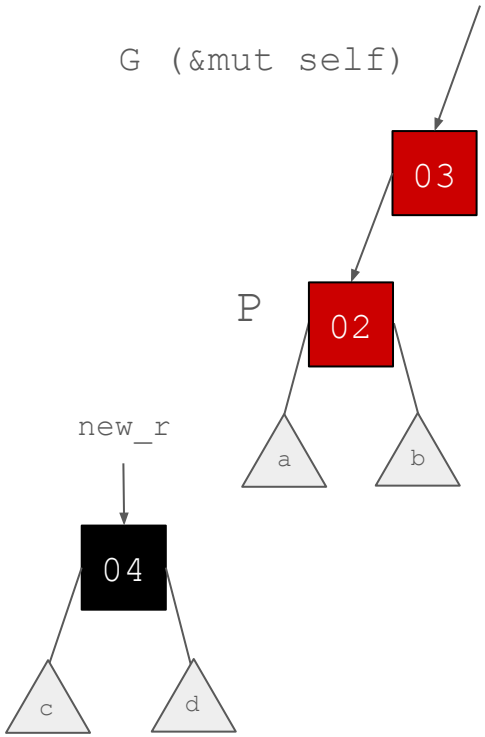

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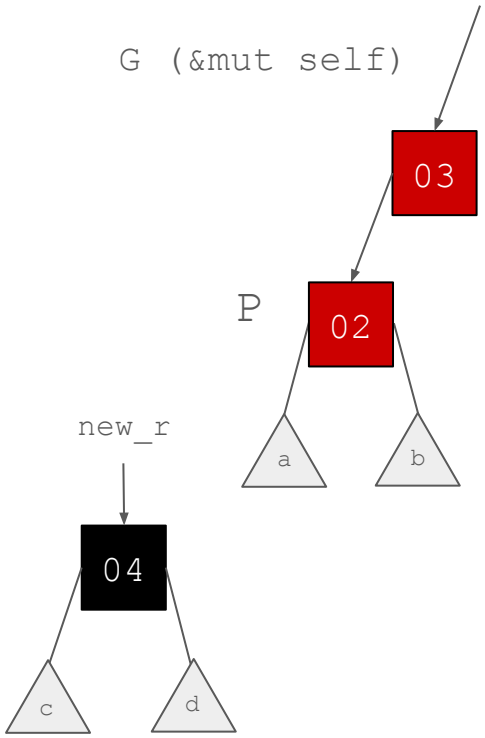
7. Attach new right child



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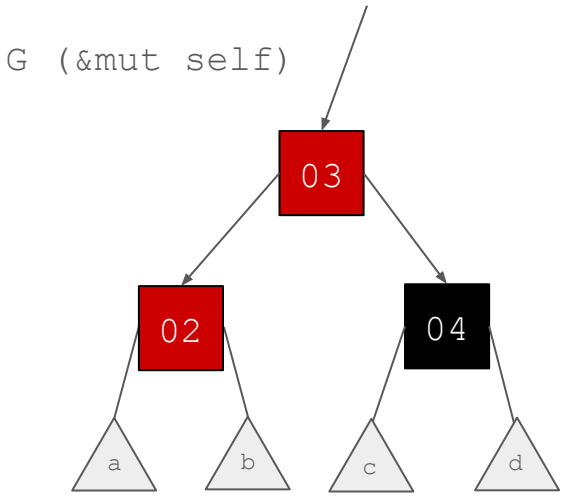

…and we are done!

Red-Black Trees: *Rust-Specific Stuff*

We give you an `for_each` which executes a function `f` on each node.

```rust
fn for_each(&self, order: &Traversal, f: &dyn Fn(&Node<K, V>)) {
    match order {
        Traversal::Preorder => {
            f(self);
            if let Some(left) = &self.left {
                left.for_each(order, f);
            }
            if let Some(right) = &self.right {
                right.for_each(order, f);
            }
        }
         ...
    }
}
```

Really helpful for writing tests, also look at the `check_tree` helper function to check invariants!